ARTICLE | Article in Press

Multi-level malware subfamily attribution from RGB executable images using attention-augmented deep learning

Maheep Saxena · Tanurup Das

Received: 22 July 2026 / Accepted: 5 October 2026
© The Author(s) 2026

Abstract

Malware analysis in modern cybersecurity increasingly requires decisions at multiple levels of granularity, including maliciousness detection, malware family identification, and fine-grained subfamily attribution. While image-based malware classification has demonstrated promising performance for binary detection and family-level recognition, the application of RGB executable image representations to hierarchical malware subfamily attribution remains largely unexplored. This study presents a unified hierarchical deep learning framework for malware classification using RGB representations of Windows Portable Executable (PE) files, enabling simultaneous malware detection, family classification, and 33-subfamily attribution within a multi-output architecture. Four attention-augmented transfer learning models ResNet50V2-CBAM, DenseNet169-CBAM, Xception-NonLocal, and VGG19-Triplet Attention were evaluated under identical datasets, taxonomy, preprocessing, and training conditions to ensure fair comparison. Experimental results achieved 99% binary classification accuracy, 96–97% family-level accuracy, and 91–95% subfamily-level accuracy, with macro F1-scores up to 0.89 for fine-grained attribution. A comprehensive experimental evaluation using per-class precision, recall, F1-score, and confusion-matrix analysis showed closely comparable performance among the three strongest attention-based models, with the fully fine-tuned VGG19 variant trailing on the most difficult variants. A protocol-matched comparison with our previously published grayscale framework further showed that RGB executable images provide competitive hierarchical performance while enabling direct compatibility with modern three-channel transfer learning architectures. This study represents one of the first systematic evaluations of RGB executable image representations for hierarchical malware subfamily attribution across a 33-subfamily taxonomy. The proposed framework provides an effective and computationally efficient approach for fine-grained malware attribution, supporting practical applications in digital forensics, cyber threat intelligence, and security operation centers.

Keywords Malware subfamily attribution · Hierarchical malware classification · RGB executable images · Transfer learning · Attention mechanisms · Windows portable executable (PE) files · Digital forensics

This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.



https://doi.org/10.1038/s41598-026-75054-6

Published online: 16 October 2026

scientific reports

Introduction

Malware defence has three questions to answer about every suspicious file, and they are not equally hard. The first, is this file malicious, is largely solved for known threat classes: modern learned detectors routinely report binary accuracies above 99%[1–3]. The second, which family does it belong to, has been the principal focus of two decades of research and now also exceeds 95% on the standard benchmarks[4,5]. The third question, which specific variant or subfamily is this, is the one incident responders actually need for campaign correlation, infrastructure attribution, and forensic case linkage, and it is the question that has received the least attention in the literature. Distinguishing the Akira ransomware lineage from Rhysida, or the AZORult stealer from PureLogStealer, requires separating binaries that share loaders, packers, crypters, and substantial code, differences far subtler than those that separate a trojan from benign software[6,7]. The present paper addresses this third question. Its subject is not image conversion for its own sake but fine-grained, hierarchical malware attribution: a single framework that answers all three questions jointly, evaluated at a granularity of 33 subfamilies across five families, which is substantially finer than the 9-family Microsoft and 25-family Malimg taxonomies on which most image-based results are reported[4,8]. The representational vehicle is an RGB encoding of the raw executable, chosen because colour representations have shown consistent family-level advantages over grayscale[5,9] yet have not been evaluated at variant granularity, and because three-channel inputs permit the direct use of ImageNet-pretrained backbones and their attention extensions without architectural modification. Four attention-augmented models are trained and evaluated under strictly identical conditions, and a fifth, combining channel attention with a sequence head, is subsequently evaluated under the same protocol to test whether the outcome of that comparison reflects the representation or the architectural choices; their differences are examined through a detailed per-class and confusion-matrix analysis rather than headline accuracy alone, and the whole framework is compared, on the same taxonomy, sampling design, and evaluation protocol, against our previously published grayscale framework[10], giving the field a controlled RGB versus grayscale data point at the variant level that has so far been missing.

The scope of the claimed contribution is stated explicitly. RGB byte-packing, ImageNet transfer learning, and the CBAM, non-local and triplet attention modules are each established techniques, and we do not propose a new attention operator. The contribution lies in the research problem addressed and in the controlled experimental design. This is summarized in Table 16, 27 of the 28 comparator studies we identified report no variant-level metric at all, and every RGB study among them stops at binary or family classification. The present work is therefore, to our knowledge, the first evaluation of RGB executable images for hierarchical attribution across 33 subfamilies, the first controlled comparison of three attention families under a single malware-imaging protocol, and the first protocol-matched RGB-versus-grayscale comparison at variant granularity. That comparison shows that four attention-augmented models reach parity with the grayscale framework, and that a fifth model combining squeeze-and-excitation channel attention with a bidirectional LSTM sequence head exceeds it at the subfamily level (95% accuracy, macro F1 0.92, against 94% and 0.90), which locates the earlier parity in architectural choice rather than in the representation (Section "Comparison with the grayscale baseline framework").

This paper is organized in the following flow: Section "Related work" reviews the relevant literature. Section "Methodology" describes the dataset, the RGB representation, the four architectures, and the training and evaluation protocol. Section "Experiments and results analysis" reports the experimental results at the binary, family and subfamily levels, together with confusion-matrix and model-complexity analyses. Section "Discussion" discusses the results in relation to published work and to our previous grayscale framework, examines the structural basis of subfamily confusions and the multi-task loss formulation, and states the limitations of the study. Section "Conclusion" concludes.

Motivation and research gap

Although malware image classification has progressed rapidly over the past decade, the majority of existing studies have concentrated on binary malware detection or malware family classification, where benchmark performance has approached saturation on widely used datasets[8,9]. From an operational perspective, however, malware analysts rarely require only a family label. Digital forensic investigations, cyber threat intelligence, and incident response increasingly depend on identifying the specific malware variant or subfamily responsible for an attack, as this level of attribution supports campaign correlation, infrastructure tracking, malware lineage reconstruction, and evidence linkage across related incidents[6,7]. Distinguishing closely related variants is considerably more challenging than separating unrelated malware families because variants frequently share executable sections, code reuse, packing techniques, and obfuscation strategies that produce highly similar structural characteristics after executable-to-image conversion. Recent advances in malware imaging have also introduced RGB executable image representations as an alternative to conventional grayscale images[5,11,12]. These representations enable direct utilization of mature ImageNet-pretrained convolutional architectures while preserving three-channel inputs compatible with modern attention mechanisms[13,14]. Nevertheless, current RGB-based studies remain focused almost exclusively on binary detection or family-level recognition[5,11,12], leaving unanswered whether RGB representations can effectively discriminate between closely related malware variants. Consequently, the principal motivation of this work is not merely to investigate RGB executable images, but to determine whether RGB representations can support fine-grained hierarchical malware attribution at the malware subfamily level while maintaining robust performance across binary detection and family classification within a unified framework.

Three principal research gaps are identified in the existing literature. First, colour-based malware image representations have been evaluated primarily for binary detection and family-level classification, with limited investigation of their effectiveness for fine-grained variant (subfamily) classification. Second, no study has performed a controlled comparison of attention mechanisms within a common malware image classification framework, limiting the understanding of their relative contribution to feature learning and classification performance. Third, hierarchical multi-output malware classification has been explored only using grayscale image representations, whereas an equivalent RGB-based hierarchical framework evaluated under an identical experimental protocol has not been reported. The present study addresses these gaps by systematically evaluating RGB image representations against a matched grayscale baseline, conducting a controlled comparison of attention mechanisms, and implementing a hierarchical multi-output classification framework spanning binary, family, and variant-level malware classification.

Objectives and contributions

To investigate whether RGB executable image representations can effectively support fine-grained hierarchical malware attribution, this study evaluates four attention-augmented transfer learning architectures that share a common classifier design but incorporate different pretrained backbones and attention mechanisms to enhance feature representation at multiple spatial and semantic levels, together with a fifth architecture that combines channel attention with a bidirectional sequence head. Specifically, the proposed framework investigates:

- **ResNet50V2 integrated with the convolutional block attention module (CBAM),** which sequentially refines channel and spatial feature responses to improve the localization of discriminative malware characteristics.
- **DenseNet169 integrated with the convolutional block attention module (CBAM),** which combines dense feature propagation with channel and spatial attention to enhance feature reuse and hierarchical representation learning.

- **Xception integrated with a non-local self-attention block,** which models long-range spatial dependencies by computing pairwise relationships between all feature locations, thereby capturing global contextual information within malware images.
- **VGG19 integrated with triplet attention,** which models cross-dimensional interactions between channel and spatial features through lightweight rotational attention branches while maintaining computational efficiency.
- **ResNet101V2 integrated with a squeeze-and-excitation block and a bidirectional LSTM head (M5),** which recalibrates channel responses and then models the 49 spatial positions of the final feature map as a sequence read in both directions, combining channel attention with the sequence modelling used in our grayscale framework.

All five architectures are optimized under an identical hierarchical multi-output learning framework, enabling a controlled comparison in which the backbone and attention mechanism constitute the primary experimental variables, while the dataset, preprocessing pipeline, optimization strategy, loss functions, and evaluation protocol remain unchanged. From a single RGB executable image, each network simultaneously predicts (i) binary classification (benign or malware); (ii) malware family classification across five major malware families; and (iii) fine-grained malware subfamily attribution across 33 malware subfamilies.

The main contributions of this work can be summarised as follows:

I. Problem-level contribution. We demonstrate that hierarchical malware classification across 33 malware subfamilies is feasible using RGB representations of Windows PE files, enabling binary malware detection, malware family identification, and fine-grained subfamily attribution without relying on additional static or dynamic features during inference. To our knowledge, no prior peer-reviewed study has evaluated RGB executable images below family-level granularity (Table 16).
II. Controlled-comparison contribution. We present a unified and reproducible hierarchical multi-output framework that jointly performs binary, family, and subfamily classification while enabling a controlled comparison of five attention-augmented transfer learning architectures under identical datasets, pre-processing pipelines, optimization strategies, and evaluation protocols. We find that three structurally different attention mechanisms applied to a pooled feature map converge on closely comparable subfamily performance (macro F1 0.88–0.89), whereas the addition of a bidirectional sequence head over the same feature map (M5) raises subfamily accuracy to 95% and macro F1 to 0.92, indicating that the manner in which the final feature map is aggregated, rather than the attention design, is the principal architectural determinant at this granularity.
III. Representation-level contribution. We provide a protocol-matched comparison between the proposed RGB framework and our previously published grayscale framework[10] using the same hierarchical taxonomy and sampling design, thereby isolating the influence of the input representation. This comparison shows that the four pooled-attention RGB models match the grayscale framework at the subfamily level (94% accuracy; macro F1 0.89 versus 0.90), and that M5 exceeds it (95%; macro F1 0.92), the first image-only model to do so on this taxonomy; it also shows that an almost identical cluster of variants resists attribution under both representations and all eight models, which locates the residual difficulty in the malware's shared code lineage rather than in the encoding (Sections "Comparison with the grayscale baseline framework" and "Structural basis of subfamily confusions").
IV. Through comprehensive experimental evaluation, including per-class metrics, confusion-matrix analysis and model-complexity measurement, we support the practical applicability of the framework to digital forensic investigations and cyber threat intelligence workflows, and identify the 13.8 M-parameter DenseNet169-CBAM configuration as an efficient deployment default.

Related work

Image-based malware analysis and the grayscale line

Nataraj et al.[8] established the executable-as-image paradigm, showing that GIST texture features over grayscale byte plots separate 25 families at roughly 98% accuracy. Deep convolutional models then absorbed the feature engineering: Kalash et al.[4] reached 98.52% on Malimg and 99.97% on the Microsoft corpus, and Cui et al.[15] showed a compact CNN suffices at 94.5%. Critically examined, this line of work demonstrates two things. It demonstrates that global byte texture is highly family-discriminative, and it demonstrates saturation: the benchmark taxonomies are coarse, a decade or more old, and headroom at the family level is measured in fractions of a point. What the grayscale line does not demonstrate is behaviour where texture stops being distinctive, namely among sibling variants that share most of their bytes. The saturation of family benchmarks has, we argue, systematically diverted the field away from the harder attribution problem rather than toward it.

RGB and multi-channel representations

Colour entered the field through two distinct routes that are often conflated. The first packs raw bytes into three channels with no added semantics; Vasan et al.[5] showed such colour inputs outperform grayscale under a fine-tuned CNN, reaching 98.82% on Malimg. The second engineers each channel: Zhao et al.[9] fused ASCII, entropy, and hexadecimal planes into an RGB composite and reported 99.99% on the Microsoft dataset, and dual-stream designs process colour and grayscale views independently before fusing decisions[11]. Transfer learning studies with colour and RGBA inputs report similar family-level strength[12]. The critical observation is that every one of these results is a binary or family result on a coarse taxonomy. Three structural reasons explain why RGB studies stop there. First, the public benchmarks that make comparison possible simply lack subfamily labels, and building a balanced variant-level corpus requires per-sample static and dynamic validation at a curation cost most studies avoid. Second, family-level gains are easy to show because colour adds information density where texture is already discriminative; at variant level the risk that siblings share texture and colour alike is far more threatening to a positive result. Third, evaluation burden grows quadratically with granularity: a 34-class confusion structure demands per-class reporting and detailed confusion analysis that a 9-class study can skip. The consequence is a genuine gap: the colour literature asserts representational superiority without ever testing it where representation matters most.

Relative to these RGB studies, the present work differs in three specific respects. (a) Granularity: Vasan et al.[5], Zhao et al.[9], Liaqat et al.[11] and Younas et al.[12] evaluate 9–25 families or a binary decision; we evaluate 33 subfamilies nested within five families plus benign, jointly with the coarser levels. (b) Encoding: we use naive byte-triplet packing (Section "RGB representation of executables") rather than engineered entropy or ASCII channels, precisely so that any benefit can be attributed to the three-channel input itself and to pretrained-filter compatibility rather than to hand-designed features. (c) Baseline: none of the cited RGB studies compares against a grayscale system trained on the same variant-level taxonomy under the same protocol; we do (Section "Comparison with the grayscale baseline framework"). We do not claim that RGB packing, transfer learning or the attention modules are new; we claim that their controlled evaluation in this setting answers an open question that the family-level literature has left untested.

Hierarchical classification and variant attribution

Hierarchy-aware malware systems exist but mostly operate on features rather than images and stop above the variant level. Alsaedi et al.[16] combine hierarchical deep learning with a doubly regularized loss over static PE features, categorizing 14 families; Bao et al.[17] use hierarchical attention over assembly and API views for interpretable family classification; Hussain et al.[7] classify ransomware family-wise; and attribution-oriented work

such as the TCN-GAN of Chen and Yan[18] targets APT groups, an organizational rather than code-lineage notion of identity. The only published image-only system evaluated on a 33-subfamily taxonomy with joint binary, family, and variant outputs is our own grayscale framework[10]. The gap here is precise: hierarchical designs exist, variant taxonomies exist, and image representations exist, but their intersection, hierarchical variant attribution from images alone, has a single data point, and it is grayscale. Whether colour changes that picture is an open empirical question this study answers.

Attention mechanisms and transfer learning in malware imaging

Attention has entered malware imaging piecemeal. Yakura et al.[19] attached attention maps to a CNN over binary images, gaining both accuracy and analyst-facing localization; Wang et al.[20] built an IoT detector on a CBAM-derived module paired with DenseNet; Kim et al.[21] applied cross-modal attention to non-disassembled files; and Lu et al.[22] showed self-attentive models capture long-range structure in raw bytes at real-time cost. On the architectural side, the underlying mechanisms are well characterized in vision: CBAM factorizes attention into sequential channel and spatial stages[13], non-local blocks compute full pairwise positional affinities[23], and triplet attention couples channel and spatial dimensions through rotated branches at near-zero parameter cost[14]. Squeeze-and-excitation blocks[24] recalibrate channel responses from a global descriptor and are the simplest channel-attention form; bidirectional LSTMs[25] read a feature sequence in both directions and, applied to flattened grayscale feature maps, formed one of the three heads of our previous framework[10]. The critical weakness of the malware literature here is comparability: each mechanism is reported on a different dataset, taxonomy, backbone, and training regime, so the field cannot say which attention design suits a given representation. Controlled comparisons of the kind routine in vision, same data, same losses, same schedule, varied module, are absent for malware imagery. That absence motivates the design of this study, in which the attention module and backbone are the only manipulated variables.

Transformers and the interpretability context

Vision transformers[26] have reached malware imaging through hybrids: Ashawa et al.[27] pair ResNet-152 with a ViT and report 99.62% family accuracy on grayscale images, Jo et al.[28] exploit ViT attention for Android bytecode, and dual-attention transformers over API sequences reach 96.06%[29]. Capsule networks form a parallel structural line[30–32]. Two critical points follow. First, transformer gains at family level come with quadratic attention cost and heavier deployment footprints, and none of these studies evaluates below family granularity, so their advantage where it matters for attribution is unestablished. Second, the attention modules used in the present framework are themselves partially interpretable: CBAM spatial masks and triplet-attention planes are directly inspectable, and the family and subfamily confusion structure reported below localizes error to specific, structurally motivated variant clusters rather than treating misclassification as noise. This gives the framework an interpretable account of its own failures without requiring additional post-hoc machinery.

Multi-task loss weighting in hierarchical classification

Multi-output networks are commonly trained on an unweighted sum of per-task losses, but the adequacy of equal weighting depends on the relative difficulty, class cardinality and convergence rate of the tasks. Kendall et al. proposed learning task weights from homoscedastic uncertainty[33], and GradNorm balances tasks by equalizing gradient magnitudes during training[34]. In malware imaging, the hierarchical systems cited above[10,16] all use fixed (typically equal) weights, and none reports an ablation of the weighting policy. Section "Rationale for equal task weighting" states the rationale for the equal weighting adopted here and Section "Multi-task loss formulation" examines its consequences and the conditions under which alternatives could be expected to differ.

Methodology

This study adopts a static malware analysis framework in which Windows Portable Executable (PE) files are transformed into RGB images and subsequently classified using a hierarchical multi-output deep learning architecture. The proposed framework simultaneously performs three classification tasks from a single RGB executable image: binary malware detection, malware family identification, and fine-grained malware subfamily attribution. Unlike conventional malware analysis pipelines that rely on handcrafted static features or dynamic behavioural analysis during deployment, the proposed framework requires only the RGB image representation of the executable at inference time. Static and dynamic analyses are performed exclusively during dataset construction to establish reliable hierarchical ground-truth labels, whereas all prediction tasks during inference are based solely on the learned visual characteristics of the executable image.

To ensure a fair and reproducible evaluation, identical datasets, hierarchical taxonomies, preprocessing procedures, optimization strategies, and evaluation protocols are employed for all investigated architectures. Consequently, the backbone and attention mechanism constitute the principal experimental variables, allowing an objective comparison of different attention-augmented transfer learning models under controlled conditions. The overall workflow of the proposed framework is illustrated in Fig. 1. This section is confined to the description of the data, representation, architectures and training protocol; experimental observations are reported in Section "Experiments and results analysis" and interpreted in Section "Discussion".

Dataset and sample labelling

The corpus comprises Windows PE binaries from publicly available research repositories (for example MalwareBazaar and VirusShare) and a curated benign set of system files and trusted applications. Malicious samples span five families (Adware, Trojan, Spyware, Rootkit, Ransomware) across 33 subfamilies, collected through May 2025 so that both mature strains and recent campaigns are represented. Each malicious sample was examined

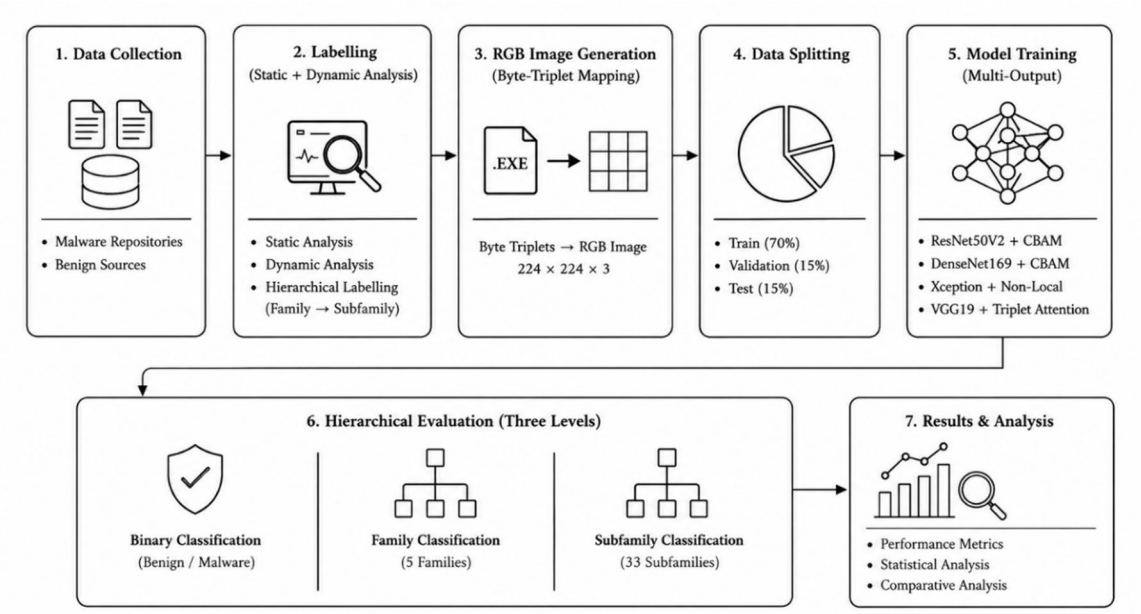


Fig. 1 End-to-end workflow of the hierarchical RGB malware attribution framework: sample collection, two-level static and dynamic labelling, byte-triplet RGB conversion, balanced partitioning, and multi-output training and evaluation of four attention-augmented models.

statically (strings, imported APIs, section sizes, section-wise entropy) and dynamically in a sandbox (file system changes, registry edits, network activity, persistence attempts); the combined evidence assigned one family label and one subfamily label per sample. Benign files carry the label Bening at both levels. Class names in all tables reproduce the label strings of the experimental pipeline verbatim so that tables and confusion matrix figures agree exactly.

RGB representation of executables

Mathematical formulation

Let a Windows Portable Executable (PE) file be represented as a sequence of bytes $B = \{b_0, b_1, \ldots, b_{L-1}\}$, where $b_i \in \{0, \ldots, 255\}$ denotes the ith byte and L is the total file length. The byte stream is partitioned into consecutive triplets, each forming one RGB pixel, so that $K = \lceil L/3 \rceil$ pixels are produced. The kth pixel is defined as $P_k = (R_k, G_k, B_k) = (b_{3k}, b_{3k+1}, b_{3k+2})$, where missing bytes in the final triplet, if any, are zero-padded. The resulting pixel sequence is arranged sequentially into a two-dimensional image following the predefined preprocessing strategy. The RGB image is resized to 224×224 pixels using bilinear interpolation and normalized to [0, 1], producing an input tensor of size $224 \times 224 \times 3$.

RGB conversion algorithm

1. Read the PE file as a raw byte stream. 2. Partition the byte stream into consecutive groups of three bytes. 3. Zero-pad the final triplet if required. 4. Map each triplet directly to the R, G and B channels of one pixel. 5. Arrange the pixels into a two-dimensional image. 6. Resize the image to 224×224 using bilinear interpolation. 7. Normalize pixel values by dividing by 255. The conversion requires only a single sequential scan of the executable and does not require disassembly, entropy estimation, opcode extraction or section-wise parsing.

Computational complexity

The executable-to-RGB conversion requires a single sequential pass over the byte stream, resulting in O(L) time complexity, where L is the executable size in bytes. Memory complexity is $O(K) = O(L/3)$, excluding the fixed memory required for the resized image. Compared with engineered multi-channel approaches requiring entropy estimation, opcode extraction or Markov statistics, the proposed representation incurs minimal preprocessing overhead and is therefore suitable for large-scale malware triage.

Rationale for the RGB representation

Conventional grayscale malware imaging maps one byte to one pixel. In contrast, the proposed RGB representation groups three consecutive bytes into a single RGB pixel, preserving neighbouring byte relationships across three colour channels while maintaining compatibility with ImageNet-pretrained CNNs. This enables direct use of transfer learning and attention mechanisms without modifying the network input. The representation remains computationally efficient; however, it is sensitive to byte insertions or deletions that alter channel alignment. Furthermore, the RGB channels do not explicitly distinguish executable code, metadata or data sections, and resizing large executables to a fixed resolution inevitably introduces information loss. Figure 2 illustrates the complete RGB conversion pipeline.

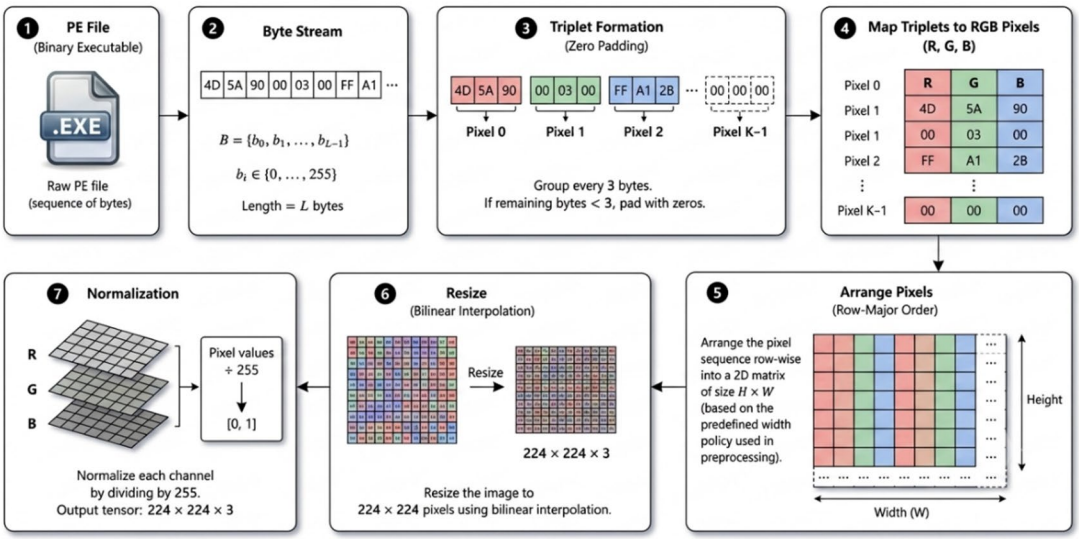


Fig. 2 Byte-triplet RGB conversion. Consecutive bytes fill the red, green, and blue channels of successive pixels; the pixel sequence is arranged row-major, resized to 224 × 224 with bilinear interpolation, and normalized to [0, 1].

Table 1 Distribution of data across training, validation, and testing partitions.

Class type	Subfamilies	Training	Validation	Testing
Malware	33	9900	1650	1650
Benign (Bening)	None	9900	1650	1650
Total		19,800	3300	3300

Table 2 Family and subfamily composition of the malware portion of the dataset.

Family	Subfamilies	Test samples	Share of malware test set (%)
Adware	5	250 (5 × 50)	15.2
Ransomware	10	500 (10 × 50)	30.3
Rootkit	4	200 (4 × 50)	12.1
Spyware	5	250 (5 × 50)	15.2
Trojan	9	450 (9 × 50)	27.3
Malware total	33	1,650	100

Dataset splitting

Each of the 33 subfamilies contributes 300 training, 50 validation, and 50 test images; benign samples are split identically at matching scale, so the test partition holds 3300 images (1650 malware, 1650 benign). Tables 1 and 2 summarize the design. Subfamily-level balance plus identical splits across models ensures that performance differences are attributable to architecture. Because subfamilies per family are uneven (Ransomware ten, Rootkit four), family-level support differs even under perfect subfamily balance; this hierarchical imbalance drives the macro versus weighted F1 divergence analysed in Section "Subfamily attribution".

Data processing and label encoding

Labels are one-hot encoded for three outputs: y1 binary (2 classes), y2 family (6 classes, five families plus benign), y3 subfamily (34 classes, 33 subfamilies plus benign). Final tensors are images X of shape (N, 224, 224, 3) with label vectors (y1, y2, y3) used jointly in the multitask objective.

Table 3 Architecture summary of the five attention-augmented models.

Model	Backbone	Attention module	Fine-tuning policy	Parameters
M1	ResNet50V2 (ImageNet)	CBAM (channel + spatial)	Backbone frozen, then last 50 layers unfrozen	25.2 M
M2	DenseNet169 (ImageNet)	CBAM (channel + spatial)	Backbone frozen, then last 50 layers unfrozen	13.8 M
M3	Xception (ImageNet)	Non-local block (embedded Gaussian)	Backbone frozen, then last 50 layers unfrozen	29.8 M
M4	VGG19 (ImageNet)	Triplet attention	Backbone frozen, then last 50 layers unfrozen (all backbone layers, since VGG19 has fewer than 50 layers)	20.2 M
M5	ResNet101V2 (ImageNet)	Squeeze-and-excitation block + BiLSTM(256) sequence head	Backbone frozen, then last 50 layers unfrozen	≈ 47.9 M

All models share the two-phase fine-tuning policy detailed in Table 4.

Table 4 Shared two-phase training configuration.

Setting	Value
Framework	TensorFlow/Keras
Input	224 × 224 × 3, normalized to [0, 1]
Optimizer	Adam
Phase 1 (frozen backbone)	Learning rate 1e−4, up to 25 epochs
Phase 2 (fine-tuning)	Last 50 backbone layers unfrozen, learning rate 1e-5, up to 10 epochs
Batch size	32
Callbacks	EarlyStopping (val_loss, patience 5, restore best weights); ReduceLROnPlateau (factor 0.5, patience 3, min lr 1e−7)
Loss	Unweighted sum of three categorical cross-entropy terms (binary, family, subfamily); see Section "Rationale for equal task weighting"
Initialization	ImageNet weights
Test partition	3300 images (1650 malware across 33 subfamilies, 1650 benign)

Model architectures

Each model pairs an ImageNet-pretrained backbone with an attention module after the final convolutional stage, followed by a shared classifier stack: global average pooling, dropout 0.5, dense 256 ReLU, dropout 0.4, dense 128 ReLU, and three softmax heads of 2, 6, and 34 units. Attention was selected over sequence heads and full transformers on substantive and operational grounds: the central question concerns channel-borne signal, which channel and cross-dimension attention address directly, and attention modules add small, fixed cost to a single forward pass, unlike quadratic self-attention stacks[22,26]. Table 3 summarizes the configurations; Table 4 lists the shared two-phase training schedule.

Following the evaluation of M1-M4, several additional backbone-attention configurations were trained under the identical protocol of Table 4 to test whether the parity observed with the grayscale framework (Section "Comparison with the grayscale baseline framework") reflected the representation or the architectural choices. Only the configuration that exceeded M1-M3 at every hierarchy level is reported, as M5. M5 retains the ImageNet-pretrained backbone and the three softmax heads of the other models but replaces global average pooling with a bidirectional LSTM that reads the 49 spatial positions of the final feature map as a sequence, so that the aggregation of the feature map, rather than only its re-weighting, becomes an experimental variable (Section "ResNet101V2 with squeeze-and-excitation and a BiLSTM head (M5)").

The selected backbones represent complementary convolutional learning strategies. ResNet50V2 employs residual learning, DenseNet169 promotes dense feature reuse, Xception utilizes depthwise separable convolutions for computational efficiency, and VGG19 provides a deep and well-established plain convolutional architecture for transfer learning. Three attention mechanisms were investigated: CBAM, which sequentially applies channel and spatial attention; Non-Local Self-Attention, which models long-range spatial dependencies; and

Triplet Attention, which captures cross-dimensional interactions between channel and spatial features. Maintaining an identical classifier across all models ensures that observed performance differences arise primarily from the backbone architecture and attention mechanism. Each mechanism operates on the final feature map at a different granularity, and Section "Representation of localized discriminative features across architectures" relates these differences to the localized, high-frequency character of the cues that separate sibling subfamilies.

ResNet50V2 with CBAM (M1)

The identity-mapping residual backbone[35] yields a $7 \times 7 \times 2048$ feature map that enters CBAM[13] in two sequential stages. Channel attention pools the map by global average and global maximum, passes both descriptors through a shared two-layer perceptron with reduction ratio 8 (2048 to 256 to 2048), sums, applies a sigmoid, and reweights the channels; for a byte-triplet image this stage directly arbitrates among chromatic byte-phase features. Spatial attention then pools the reweighted map channel-wise by average and maximum, concatenates the two planes, and applies a 7×7 sigmoid convolution to produce a positional mask highlighting regions such as packed segments or import structures. The refined map follows the shared stack to the three heads (Fig. 3). Attention over binary images improves both accuracy and analyst-facing localization[19], and CBAM-derived modules are established in malware family classification[20].

DenseNet169 with CBAM (M2)

DenseNet169 concatenates every preceding layer, so its final $7 \times 7 \times 1664$ map aggregates shallow high-frequency texture, which carries fine byte-phase patterns, alongside deep semantics[36]; CBAM then arbitrates among them explicitly, with a channel MLP of ratio 8 (1664 to 208 to 1664) and the same 7×7 spatial stage (Fig. 4). The DenseNet plus channel-spatial attention pairing has direct malware precedent[20] and, at 13.8 M parameters, is the smallest model compared.

Xception with a non-local block (M3)

Convolutional receptive fields relate positions only through local composition, yet a binary can place related artefacts, an unpacking stub and its payload for instance, at opposite ends of the file and hence of the image. The non-local block[23] computes, for every position, a response weighted over all positions by pairwise affinity. In the embedded Gaussian form used here, three 1×1 convolutions project the Xception output[37] into 1024-dimensional query, key, and value spaces; softmax-normalized query-key affinities aggregate the values; and a final 1×1

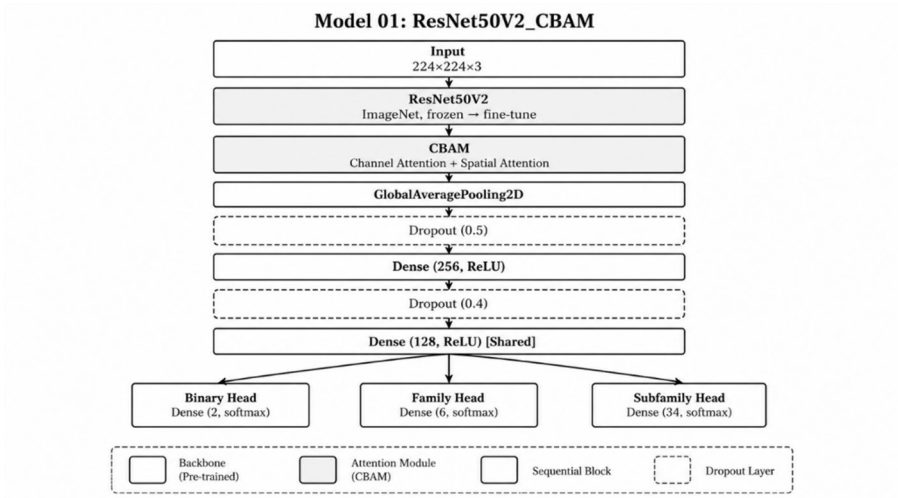


Fig. 3 ResNet50V2-CBAM (M1): backbone, CBAM with channel reduction ratio 8 and 7×7 spatial attention, shared classifier stack, and the three softmax heads (2, 6, 34).

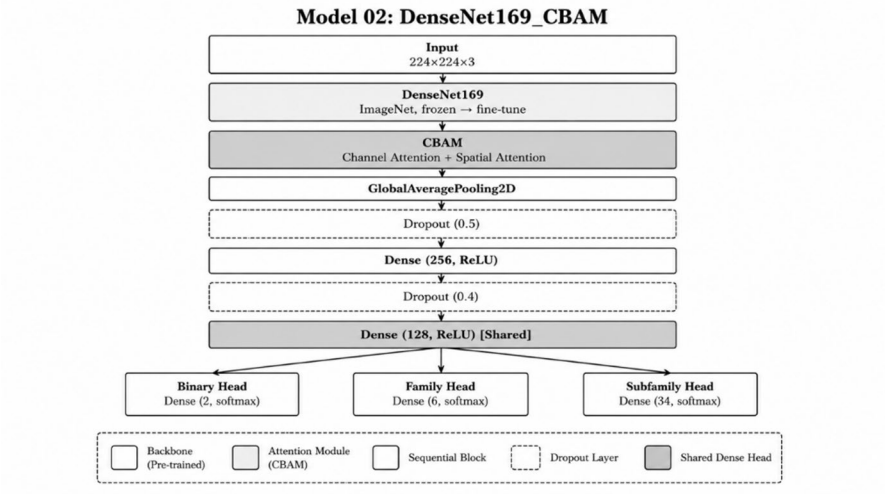


Fig. 4 DenseNet169-CBAM (M2): dense backbone, CBAM (1664 to 208 to 1664 channel MLP, 7 × 7 spatial attention), shared classifier stack, three softmax heads.

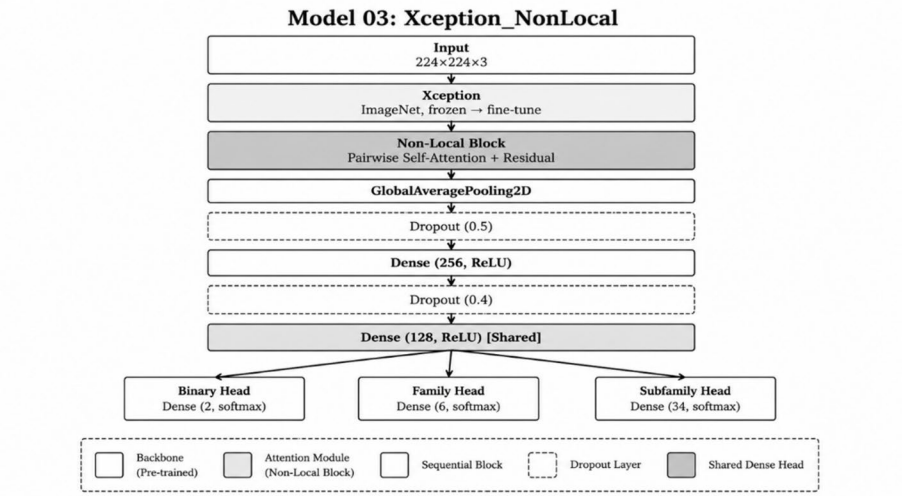


Fig. 5 Xception-NonLocal (M3): separable-convolution backbone, embedded Gaussian non-local block (1 × 1 query, key, value projections of 1024 channels, softmax affinity, restoration to 2048 channels, residual addition), shared classifier stack, three heads.

convolution restores 2048 channels before residual addition (Fig. 5). Depthwise-separable convolution keeps the added global-attention cost affordable[37]. Self-attentive models are validated on raw-byte and cross-modal malware inputs[21,22]; to our knowledge this specific block had not previously been benchmarked on malware imagery, which this study addresses.

VGG19 with triplet attention (M4)

The fourth model tests whether a plain homogeneous backbone[38], fully fine-tuned so that every layer adapts to executable statistics, can match the elaborated designs when given a cross-dimension attention head. Triplet attention[14] builds three rotated branches capturing channel-height, channel-width, and height-width interaction; each branch compresses its leading dimension by stacked average and max pooling, applies a 7 × 7 sigmoid convolution, and rotates back, and the averaged branches reweight the 7 × 7 × 512 features (Fig. 6). The module is nearly parameter-free and, unlike CBAM, models channel and spatial interaction jointly, attractive where channel identity literally encodes byte position within a triplet. Triplet attention is validated on large-scale recognition[14]; we are not aware of a prior published application to malware, so the pairing is evaluated here for the first time rather than justified by domain precedent.

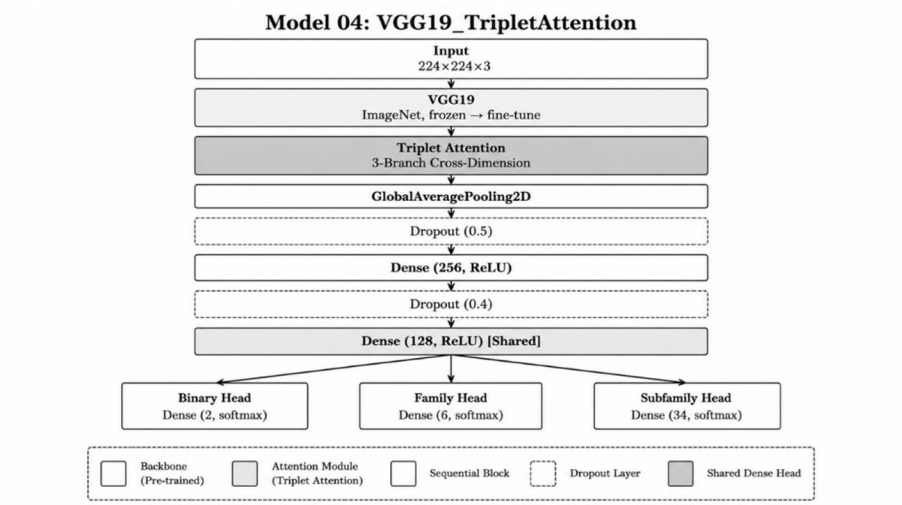


Fig. 6 VGG19-TripletAttention (M4): fully fine-tuned VGG19, triplet attention (three rotated branches with Z-pooling and 7 × 7 sigmoid convolutions, averaged), shared classifier stack, three heads.

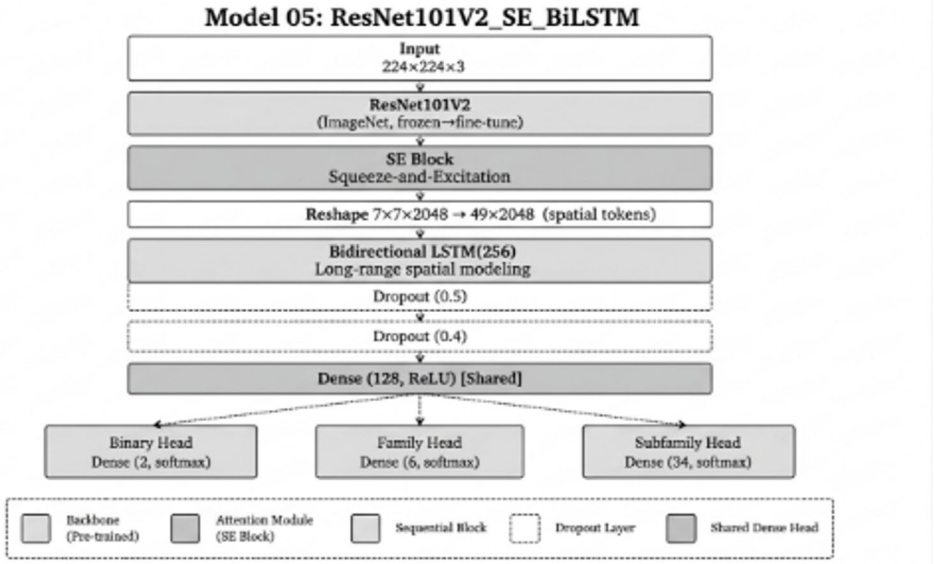


Fig. 7 ResNet101V2-SE-BiLSTM (M5): pretrained backbone, squeeze-and-excitation block, reshaping of the 7 × 7 × 2048 map into 49 spatial tokens, bidirectional LSTM (256), dropout, shared dense layer and three softmax heads.

ResNet101V2 with squeeze-and-excitation and a BiLSTM head (M5)

The fifth model tests the hypothesis, formed after the evaluation of M1–M4, that the parity of the pooled-attention models with the grayscale framework arises from the aggregation of the final feature map rather than from the RGB representation. It pairs the deeper ResNet101V2 backbone[35] with a squeeze-and-excitation block[24], which pools the 7 × 7 × 2048 map globally, passes the descriptor through a two-layer bottleneck (2048 to 128 to 2048) with sigmoid output, and rescales the channels; for a byte-triplet image this recalibrates the chromatic byte-phase features in the same manner as the channel stage of CBAM but without a spatial stage. Instead of global average pooling, the recalibrated map is reshaped into a sequence of 49 spatial tokens of 2048 dimensions and processed by a bidirectional LSTM of 256 units per direction[25], whose final states are concatenated into a 512-dimensional vector; dropout of 0.5 and 0.4, the shared dense layer of 128 units and the three softmax heads follow as in the other models (Fig. 7). The sequence head is the RGB counterpart of the CNN + BiLSTM hybrid of the grayscale

framework[10], applied to a pretrained backbone: it preserves the ordering of the 49 positions, which global pooling discards, and allows a cue at one position to be interpreted in the context of the positions that precede and follow it. M5 was trained under the two-phase schedule of Table 4.

Training configuration and hierarchical learning objective

All experiments were performed on a dedicated workstation running Windows 11 Pro and equipped with an NVIDIA RTX A1000 graphics processing unit (GPU) with 8 GB of dedicated video memory (VRAM) and CUDA 13.0 support. The proposed framework was implemented in Python 3.10 using TensorFlow 2.x with the Keras deep learning API, while cuDNN-enabled GPU acceleration was employed to accelerate model training and inference. Identical hardware, software, preprocessing, and optimization settings were maintained throughout all experiments to ensure reproducibility and facilitate an unbiased comparison of the investigated architectures.

All five attention-augmented transfer learning models were trained using an identical two-stage optimization strategy. During the first training stage, the pretrained convolutional backbone remained frozen while only the attention module and hierarchical classification heads were optimized. The models were trained for 25 epochs using the Adam optimizer with an initial learning rate of 1×10^{-4}. In the second stage, the last 50 layers of the backbone network were unfrozen and jointly fine-tuned with the classifier for an additional 10 epochs using a reduced learning rate of 1×10^{-5}, allowing the pretrained features to adapt progressively to malware-specific image characteristics while minimizing catastrophic forgetting.

Each model was trained using a mini-batch size of 32 RGB executable images. The proposed framework employs a hierarchical multi-output learning strategy, simultaneously optimizing three prediction heads corresponding to binary malware detection, malware family classification, and malware subfamily attribution. Each prediction head uses a Softmax activation and is optimized using the categorical cross-entropy loss, with the overall training objective defined as the unweighted sum of the three individual losses:

$$L_{\text{Total}} = L_{\text{Binary}} + L_{\text{Family}} + L_{\text{Subfamily}}$$

where L_{Binary}, L_{Family}, and $L_{\text{Subfamily}}$ denote the categorical cross-entropy losses associated with the three hierarchical classification tasks. To improve convergence and reduce overfitting, the shared classification head incorporated two dropout layers with dropout rates of 0.5 and 0.4, respectively. Training was monitored using the validation loss, and an EarlyStopping callback with a patience of 5 epochs was employed to terminate optimization when no further improvement was observed, while automatically restoring the best-performing model weights. Furthermore, the learning rate was adaptively adjusted using the ReduceLROnPlateau strategy, which reduced the learning rate by a factor of 0.5 after 3 consecutive epochs without improvement in validation loss, with a minimum learning rate of 1×10^{-7}.

Rationale for equal task weighting

Equal weighting of the three cross-entropy terms was adopted for three reasons. First, comparability: the grayscale predecessor[10] against which this study is protocol-matched (Section "Comparison with the grayscale baseline framework") used the same unweighted sum, and altering the objective would have confounded the representation comparison with a change in the learning problem. Second, hierarchy consistency: the three heads share a single feature vector, and an unweighted sum places no prior on which level should dominate that vector, so the learned representation is pulled toward features that are simultaneously informative at all three levels rather than toward the easiest task. Third, transparency: fixed equal weights make the objective identical across the four architectures and free of tuned hyperparameters that could favour one backbone. We acknowledge that the three tasks differ markedly in difficulty (binary accuracy saturates at 99% within a few epochs, whereas subfamily accuracy reaches 91–95%), in class cardinality (2, 6 and 34 classes), and therefore in the scale and convergence

behaviour of their losses; the subfamily term is the largest throughout training and, in effect, receives the greatest share of the gradient even under nominal equal weighting. The consequences of this choice, and the conditions under which weighted or adaptive objectives[33,34] could be expected to alter the outcome, are examined in Section "Multi-task loss formulation".

Evaluation protocol and metrics

All results are computed on the common held-out test partition of 3300 images (1650 malware across 33 subfamilies, 1650 benign), identical for all five models. For each hierarchy level we report accuracy, per-class precision, recall and F1-score, the macro-averaged F1 (equal weight to every class, and therefore the most demanding metric at the 34-class subfamily level) and the support-weighted F1. Confusion matrices are inspected at all three levels to locate rather than merely count errors. Model complexity is measured directly from the trained artefacts (parameter count and serialized size). No test data were used for model selection; early stopping and learning-rate scheduling were driven exclusively by the validation partition.

Experiments and results analysis

This section reports the experimental observations at the three hierarchy levels, followed by the confusion-matrix and model-complexity analyses. Interpretation of these observations in relation to published work, to our previous grayscale framework, and to the structural properties of the malware is deferred to Section "Discussion".

Binary detection

At the binary level all five models converge on 99% accuracy with a macro F1 of 0.99 (Table 5). The residual errors are not symmetric: they consist almost entirely of benign executables conservatively flagged as malicious rather than malware escaping detection (malware recall 0.99–1.00; benign recall 0.98–0.99 across the five models). Figure 8 shows the confusion matrices of M1-M4 and Fig. 9 that of M5, which misclassifies 31 benign and 7 malicious executables. That four backbones as different as a residual network, a densely connected network, a separable-convolution network, and a plain deep stack, paired with three unrelated attention mechanisms, all arrive at the same score is consistent with the saturation repeatedly reported for image-based detection[4,5,10]; its implications for deployment are discussed in Section "Interpretation of the RGB representation and operational implications".

Table 5 Binary classification performance on the test set (n = 3300).

Model	Class	Precision	Recall	F1-score	Support	Accuracy
ResNet50V2-CBAM	Bening	0.99	0.98	0.99	1650	0.99
	Malware	0.98	0.99	0.99	1650	
DenseNet169-CBAM	Bening	1.00	0.98	0.99	1650	0.99
	Malware	0.98	1.00	0.99	1650	
Xception-NonLocal	Bening	1.00	0.98	0.99	1650	0.99
	Malware	0.98	1.00	0.99	1650	
VGG19-TripletAttention	Bening	1.00	0.98	0.99	1650	0.99
	Malware	0.99	1.00	0.99	1650	
ResNet101V2-SE-BiLSTM	Bening	1.00	0.98	0.99	1650	0.99
	Malware	0.98	1.00	0.99	1650	

https://doi.org/10.1038/s41598-026-75054-6

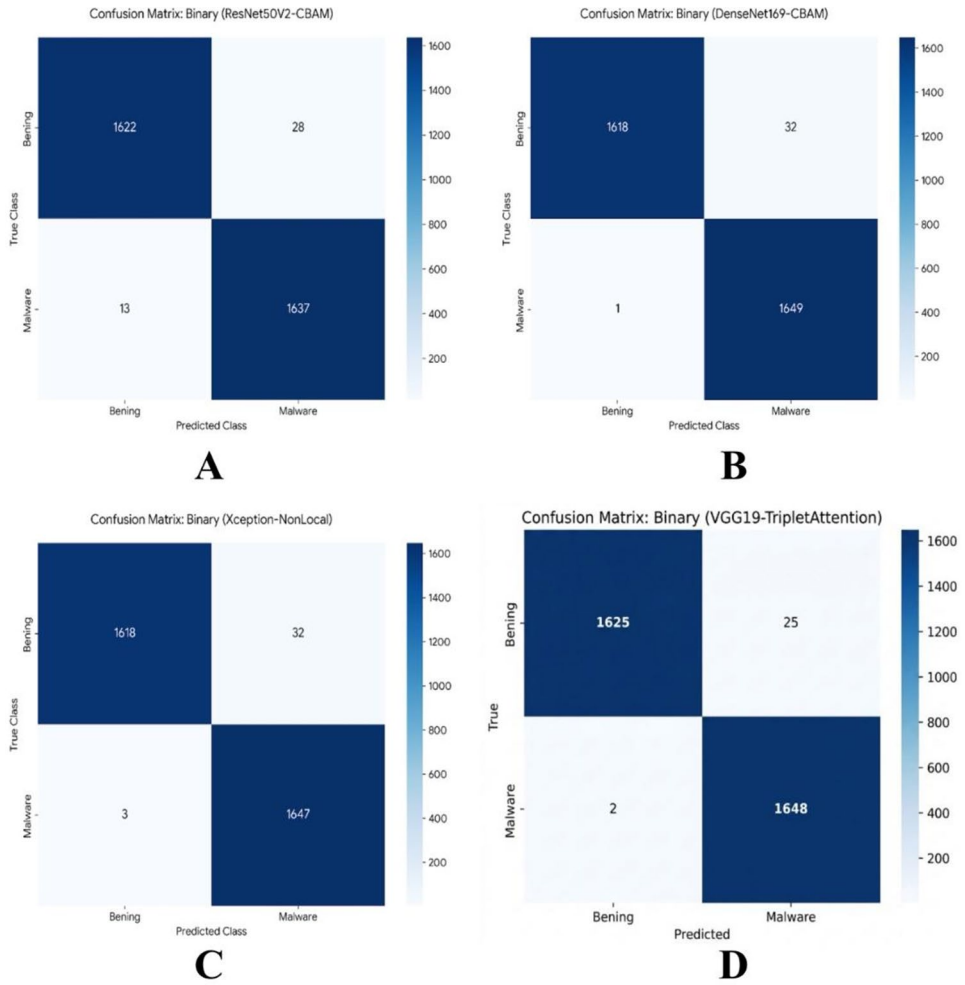


Fig. 8 Binary confusion matrices for M1 (ResNet50V2-CBAM), M2 (DenseNet169-CBAM), M3 (Xception-NonLocal), and M4 (VGG19-TripletAttention).

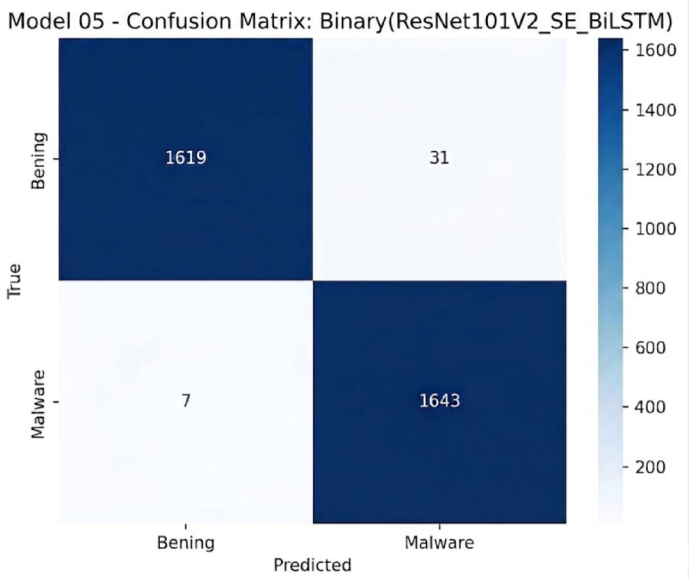


Fig. 9 Binary confusion matrix for M5 (ResNet101V2-SE-BiLSTM).

Family identification

Family identification separates the models only marginally: DenseNet169-CBAM and Xception-NonLocal reach 97% accuracy with weighted F1 of 0.97, while ResNet50V2-CBAM and VGG19-TripletAttention reach 96% (Table 6). M5 also reaches 97% and records the highest family-level macro F1 (0.96), with Adware 0.95, Spyware 0.92 and Trojan 0.94. Rootkit and Ransomware are recovered almost perfectly by every model (F1 0.96–0.99), and the benign class is likewise near-perfect (F1 0.99). Almost all residual error concentrates on the Trojan-Spyware boundary (Trojan F1 0.90–0.93; Spyware F1 0.89–0.93). The one model-specific signature is VGG19-TripletAttention, whose Adware F1 falls to 0.84 as adware samples are absorbed into Trojan. Figure 10 shows the family confusion matrices of M1–M4. The structural reasons for the Trojan-Spyware overlap and for the VGG19 Adware deficit are discussed in Sections "Family level" and "Structural basis of subfamily confusions" (Fig. 11).

Subfamily attribution

Subfamily attribution is where the models diverge and where the difficulty of the task becomes visible. Discrimination among 33 variants, with benign as a 34th class, is qualitatively harder than family classification. We hypothesize, on the basis of the confusion structure reported below and of threat-intelligence knowledge of the lineages involved, that this is because families are separated by gross architectural differences, whereas sibling subfamilies are frequently forks, rebrands, or licensed builds of a common codebase and differ mainly in localized, high-frequency detail; Section "Structural basis of subfamily confusions" sets out the supporting evidence for this interpretation and its limits. Tables 7, 8, 9, 10 and 11 give the per-subfamily metrics and Table 12 the aggregate accuracy, macro F1, and weighted F1. Among the pooled-attention models, ResNet50V2-CBAM and Xception-NonLocal lead at 94% accuracy with macro F1 of 0.89, DenseNet169-CBAM follows at 93% (macro F1 0.88), and VGG19-TripletAttention trails at 91% (macro F1 0.82). M5 exceeds all four, reaching 95% accuracy with macro F1 0.92 and weighted F1 0.95 (Table 11): 29 of the 33 subfamilies attain F1 of 0.83 or above, and no subfamily falls below 0.72. The gap between weighted and macro F1 is roughly 0.05 for the first three models but 0.08 for VGG19, showing that the weaker model does not degrade uniformly but concentrates its failures on a subset of hard variants while remaining competitive on the majority. VGG19-TripletAttention collapses almost completely on PureLogSteale (F1 = 0.03) and BackdoorGhostRat (F1 = 0.38), whereas the residual-, dense-, and separable-convolution backbones retain usable performance on the same classes (F1 0.62–0.73 and 0.77–0.85 respectively). The three architecturally richer pooled models share both their ranking and their error profile and differ from one another only marginally in aggregate F1, whereas VGG19 sits clearly below them. The gain of M5 is distributed rather than concentrated: relative to the best pooled model on each class, M5 improves F1 on 21 subfamilies, most markedly on AZORult (0.83 vs. 0.74), AgentTesla (0.84 vs. 0.82), DiskWritter (0.83 vs. 0.72), Pony (0.83 vs. 0.76), PureLogSteale (0.82 vs. 0.73), RemcosRAT (0.80 vs. 0.74) and ValleyRat (0.98 vs. 0.92), while ConnectWise (0.72; recall 0.58) remains the most difficult class for every model.

Table 6 Family-level F1-scores per class with overall accuracy, macro F1, and weighted F1.

Class	M1	M2	M3	M4	M5	Support
Adware	0.95	0.94	0.94	0.84	0.95	250
Bening	0.99	0.99	0.99	0.99	0.99	1650
Ransomware	0.97	0.98	0.97	0.99	0.98	500
Rootkit	0.97	0.96	0.96	0.96	0.97	200
Spyware	0.90	0.91	0.89	0.93	0.92	250
Trojan	0.90	0.92	0.93	0.90	0.94	450
Accuracy	0.96	0.97	0.97	0.96	0.97	3,300
Macro F1	0.95	0.95	0.95	0.93	0.96	
Weighted F1	0.96	0.97	0.97	0.96	0.97	

Class labels reproduce the experimental label strings verbatim.

https://doi.org/10.1038/s41598-026-75054-6

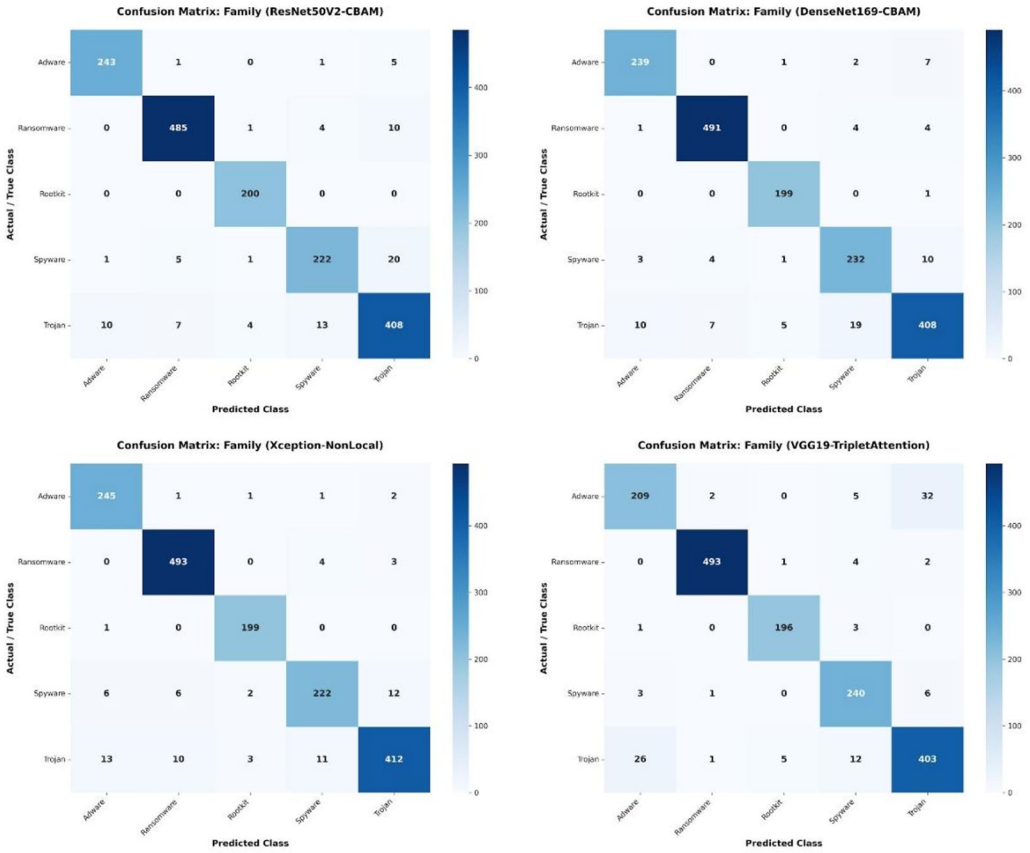


Fig. 10 Family-level confusion matrices for M1 (ResNet50V2-CBAM), M2 (DenseNet169-CBAM), M3 (Xception-NonLocal), and M4 (VGG19-TripletAttention).

Fig. 11 Family-level confusion matrix for M5 (ResNet101V2-SE-BiLSTM).

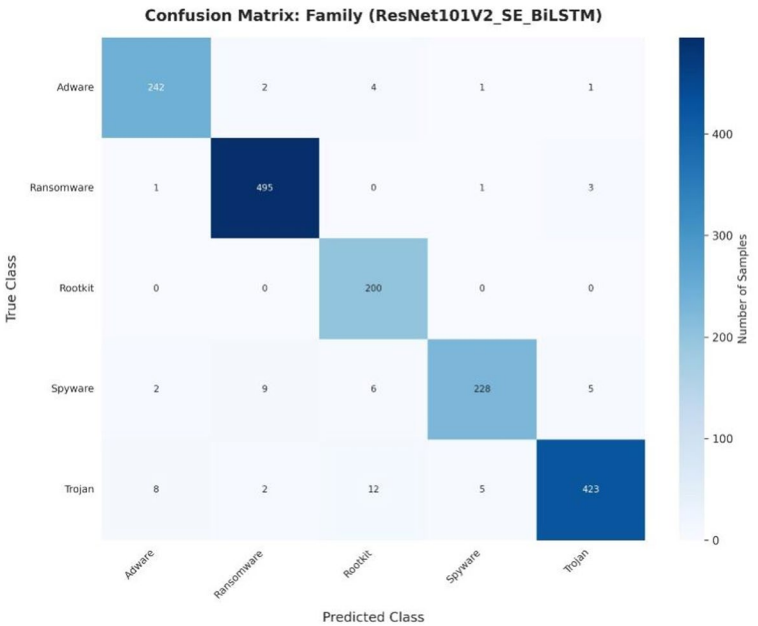


Table 7 Subfamily-wise performance for M1 (ResNet50V2-CBAM).

Subfamily	Precision	Recall	F1-score	Support
AZORult	0.61	0.88	0.72	50
Adware.Generic	0.86	0.98	0.92	50
AdwareAdload	0.94	1.00	0.97	50
AgentTesla	0.81	0.84	0.82	50
Akira	0.98	1.00	0.99	50
BackdoorGhostRat	0.92	0.70	0.80	50
Bening	0.99	0.98	0.99	1650
Berbew	0.98	1.00	0.99	50
BianLain	0.98	0.94	0.96	50
BlackMatter	0.96	0.98	0.97	50
Cactus	0.94	1.00	0.97	50
CoinMiner	0.94	0.96	0.95	50
ConnectWise	0.72	0.56	0.63	50
Cosmu	0.92	0.98	0.95	50
CypherIT	0.88	0.86	0.87	50
DarkWatchMan	0.94	0.98	0.96	50
DiskWritter	0.57	0.64	0.60	50
Expiro	1.00	0.92	0.96	50
FromBook	0.94	0.92	0.93	50
FuRootkit	0.96	1.00	0.98	50
HelloKity	0.96	0.92	0.94	50
IcedID	0.85	1.00	0.92	50
Kuping	0.94	1.00	0.97	50
Loki	0.84	0.72	0.77	50
LummaStealer	0.82	0.82	0.82	50
Medusa	0.98	1.00	0.99	50
Nitrogen	0.94	1.00	0.97	50
Pony	0.73	0.74	0.73	50
PureCrypter	0.87	0.80	0.83	50
PureLogSteale	0.82	0.64	0.72	50
RemcosRAT	0.78	0.70	0.74	50
Reworld	0.98	0.96	0.97	50
Rhysida	0.96	0.98	0.97	50
ValleyRat	0.85	0.90	0.87	50
Accuracy			0.94	3300
Macro avg	0.89	0.89	0.89	3300
Weighted avg	0.94	0.94	0.94	3300

Two features of the per-class tables are used as evidence in Section "Structural basis of subfamily confusions". First, every subfamily whose F1 falls below 0.80 in any of the five models belongs to one of two within-family groups: the commodity information stealers (AZORult, PureLogSteale, LummaStealer, AgentTesla, FromBook, all Trojan or Spyware) and the remote-access and wiper tools (RemcosRAT, DiskWritter, BackdoorGhostRat, ConnectWise). No ransomware or rootkit subfamily falls below F1 0.90 in any model. Second, the errors are directional: ConnectWise shows high precision (0.67–0.94) but low recall (0.56–0.60) in all five models, whereas AZORult shows the converse (recall 0.60–0.98, precision 0.45–0.72), a pattern that is stable across backbones, attention mechanisms and aggregation heads (Figs. 12, 13, 14, 15, 16, 17 and 18).

Table 8 Subfamily-wise performance for M2 (DenseNet169-CBAM).

Subfamily	Precision	Recall	F1-score	Support
AZORult	0.53	0.82	0.65	50
Adware.Generic	0.80	0.90	0.85	50
AdwareAdload	0.93	1.00	0.96	50
AgentTesla	0.62	0.78	0.69	50
Akira	0.98	1.00	0.99	50
BackdoorGhostRat	0.84	0.72	0.77	50
Bening	1.00	0.98	0.99	1650
Berbew	0.92	0.98	0.95	50
BianLain	0.98	0.92	0.95	50
BlackMatter	0.94	1.00	0.97	50
Cactus	0.96	1.00	0.98	50
CoinMiner	0.96	1.00	0.98	50
ConnectWise	0.85	0.56	0.67	50
Cosmu	0.93	1.00	0.96	50
CypherIT	0.82	0.98	0.89	50
DarkWatchMan	0.92	0.98	0.95	50
DiskWritter	0.76	0.68	0.72	50
Expiro	0.92	0.92	0.92	50
FromBook	0.85	0.80	0.82	50
FuRootkit	1.00	1.00	1.00	50
HelloKity	0.94	0.92	0.93	50
IcedID	0.82	0.98	0.89	50
Kuping	0.94	1.00	0.97	50
Loki	0.71	0.70	0.71	50
LummaStealer	1.00	0.80	0.89	50
Medusa	0.98	1.00	0.99	50
Nitrogen	1.00	1.00	1.00	50
Pony	0.71	0.80	0.75	50
PureCrypter	0.89	0.80	0.84	50
PureLogSteale	0.68	0.56	0.62	50
RemcosRAT	0.78	0.64	0.70	50
Reworld	1.00	0.96	0.98	50
Rhysida	0.98	1.00	0.99	50
ValleyRat	0.86	0.88	0.87	50
Accuracy			0.93	3300
Macro avg	0.88	0.88	0.88	3300
Weighted avg	0.94	0.93	0.93	3300

Confusion matrix analysis

Binary level

The binary matrices (Fig. 8) are almost purely diagonal, and the small off-diagonal mass sits overwhelmingly in the benign-to-malware cell rather than its converse. The benign files most likely to be misread are legitimate installers, updaters, and self-extracting archives, whose compressed or packed payload sections produce the same high-entropy visual texture that the models learn to associate with protected malware; the confusion is therefore driven by a genuine structural resemblance rather than by model error, and it mirrors the well-documented difficulty of separating packed benign software from packed malware in the obfuscation literature[6,39]. Because the errors fall on the conservative side, the binary head can be deployed at its native threshold as a first-stage filter, with flagged benign files resolved downstream by allow-listing rather than by relaxing the decision boundary.

Table 9 Subfamily-wise performance for M3 (Xception-NonLocal).

Subfamily	Precision	Recall	F1-score	Support
AZORult	0.67	0.82	0.74	50
Adware.Generic	0.92	0.96	0.94	50
AdwareAdload	0.91	1.00	0.95	50
AgentTesla	0.81	0.76	0.78	50
Akira	0.96	1.00	0.98	50
BackdoorGhostRat	0.89	0.82	0.85	50
Bening	1.00	0.98	0.99	1650
Berbew	0.88	0.92	0.90	50
BianLain	0.96	0.96	0.96	50
BlackMatter	0.96	0.98	0.97	50
Cactus	0.96	1.00	0.98	50
CoinMiner	0.93	1.00	0.96	50
ConnectWise	0.79	0.60	0.68	50
Cosmu	0.86	0.96	0.91	50
CypherIT	0.91	0.96	0.93	50
DarkWatchMan	0.92	0.98	0.95	50
DiskWritter	0.82	0.56	0.67	50
Expiro	0.91	0.98	0.94	50
FromBook	0.86	0.84	0.85	50
FuRootkit	1.00	1.00	1.00	50
HelloKity	0.96	0.96	0.96	50
IcedID	0.86	1.00	0.93	50
Kuping	0.94	1.00	0.97	50
Loki	0.83	0.78	0.80	50
LummaStealer	0.98	0.88	0.93	50
Medusa	0.98	1.00	0.99	50
Nitrogen	0.96	1.00	0.98	50
Pony	0.68	0.72	0.70	50
PureCrypter	0.72	0.76	0.74	50
PureLogSteale	0.66	0.82	0.73	50
RemcosRAT	0.79	0.60	0.68	50
Reworld	0.98	0.96	0.97	50
Rhysida	0.96	0.98	0.97	50
ValleyRat	0.88	0.98	0.92	50
Accuracy			0.94	3300
Macro avg	0.89	0.90	0.89	3300
Weighted avg	0.94	0.94	0.94	3300

Family level

The family matrices (Fig. 10) concentrate nearly all confusion on a single axis, Trojan against Spyware, with a secondary Adware-into-Trojan leakage that is specific to VGG19-TripletAttention. Neither pattern is arbitrary. The Trojan-Spyware overlap reflects a real ambiguity in the taxonomy itself: many modern stealers are labelled as trojans by some vendors and as spyware by others, and because both classes embed the same credential-theft and exfiltration components behind the same commodity packers, their whole-file images are legitimately similar. Errors along this axis are therefore benign for triage, since both labels route a sample into the same stealer-oriented analysis workflow, but they caution against treating the family head as authoritative for taxonomy-sensitive reporting. The Adware-Trojan leakage under VGG19 has a different origin: it is a capacity effect, in which the fully fine-tuned plain backbone fails to preserve the subtle bundler-specific texture that the residual and dense models retain, and it foreshadows the same model's larger deficit at the subfamily level. The near-absence of confusion for Ransomware and Rootkit confirms that where families differ in gross structure, the representation separates them cleanly regardless of the attention mechanism.

Table 10 Subfamily-wise performance for M4 (VGG19-TripletAttention).

Subfamily	Precision	Recall	F1-score	Support
AZORult	0.45	0.60	0.51	50
Adware.Generic	0.56	0.86	0.68	50
AdwareAdload	0.96	1.00	0.98	50
AgentTesla	0.66	0.76	0.70	50
Akira	0.96	0.98	0.97	50
BackdoorGhostRat	0.47	0.32	0.38	50
Bening	1.00	0.99	0.99	1650
Berbew	0.68	0.98	0.80	50
BianLain	0.98	0.84	0.90	50
BlackMatter	0.96	1.00	0.98	50
Cactus	0.93	1.00	0.96	50
CoinMiner	0.79	1.00	0.88	50
ConnectWise	0.67	0.56	0.61	50
Cosmu	0.86	1.00	0.93	50
CypherIT	0.79	0.98	0.88	50
DarkWatchMan	0.96	0.98	0.97	50
DiskWritter	0.67	0.40	0.50	50
Expiro	0.90	0.92	0.91	50
FromBook	0.87	0.80	0.83	50
FuRootkit	1.00	1.00	1.00	50
HelloKity	1.00	0.94	0.97	50
IcedID	0.79	0.88	0.83	50
Kuping	0.94	1.00	0.97	50
Loki	0.90	0.86	0.88	50
LummaStealer	0.89	0.82	0.85	50
Medusa	1.00	1.00	1.00	50
Nitrogen	0.98	0.96	0.97	50
Pony	0.87	0.68	0.76	50
PureCrypter	0.71	0.92	0.80	50
PureLogSteale	0.10	0.02	0.03	50
RemcosRAT	0.74	0.64	0.69	50
Reworld	1.00	0.96	0.98	50
Rhysida	0.98	0.98	0.98	50
ValleyRat	0.75	0.72	0.73	50
Accuracy			0.91	3300
Macro avg	0.82	0.83	0.82	3300
Weighted avg	0.90	0.91	0.90	3300

Subfamily level

The subfamily matrices (Figs. 12 and 13) show that error at this level is not diffuse but concentrated in a small and remarkably stable set of variants, and the per-class metrics in Tables 7, 8, 9, 10 and 11 identify its members precisely. The core is a cluster of commodity information stealers, AZORult, PureLogSteale, LummaStealer, and to a lesser extent AgentTesla and FromBook, that exchange probability mass in every model. These families circulate in the same criminal marketplaces, reuse one another's grabber modules and crypters, and are frequently distributed through the same packers, so their static images are close to genuinely overlapping. The direction of the errors is itself informative. AZORult tends to draw neighbours into itself, exhibiting high recall but low precision, whereas ConnectWise shows the opposite signature across all five models, high precision but recall as low as 0.56–0.60, indicating that a subset of its samples is systematically lost to a neighbouring class. That pattern is consistent with the nature of ConnectWise-derived malware, which weaponises a legitimate remote-administration codebase, so that much of each binary is shared, signed, benign-origin structure that provides little discriminative texture. A second overlap zone links the remote-access and wiper families RemcosRAT, DiskWritter,

Table 11 Subfamily-wise performance for M5 (ResNet101V2-SE-BiLSTM).

Subfamily	Precision	Recall	F1-score	Support
AZORult	0.72	0.98	0.83	50
Adware.Generic	0.94	0.96	0.95	50
AdwareAdload	0.93	1.00	0.96	50
AgentTesla	0.95	0.76	0.84	50
Akira	0.98	1.00	0.99	50
BackdoorGhostRat	0.88	0.84	0.86	50
Bening	0.99	0.98	0.99	1650
Berbew	0.94	1.00	0.97	50
BianLain	0.96	0.92	0.94	50
BlackMatter	0.96	0.96	0.96	50
Cactus	0.94	1.00	0.97	50
CoinMiner	0.98	1.00	0.99	50
ConnectWise	0.94	0.58	0.72	50
Cosmu	0.93	1.00	0.96	50
CypherIT	0.91	0.98	0.94	50
DarkWatchMan	0.96	1.00	0.98	50
DiskWritter	0.90	0.76	0.83	50
Expiro	0.96	0.96	0.96	50
FromBook	0.84	0.94	0.89	50
FuRootkit	0.98	1.00	0.99	50
HelloKity	0.98	0.94	0.96	50
IcedID	0.86	1.00	0.93	50
Kuping	0.94	1.00	0.97	50
Loki	0.84	0.82	0.83	50
LummaStealer	0.96	0.86	0.91	50
Medusa	0.94	1.00	0.97	50
Nitrogen	0.96	1.00	0.98	50
Pony	0.90	0.76	0.83	50
PureCrypter	0.78	0.90	0.83	50
PureLogSteale	0.81	0.84	0.82	50
RemcosRAT	0.79	0.82	0.80	50
Reworld	0.98	0.98	0.98	50
Rhysida	0.98	1.00	0.99	50
ValleyRat	0.98	0.98	0.98	50
Accuracy			0.95	3300
Macro avg	0.92	0.93	0.92	3300
Weighted avg	0.96	0.95	0.95	3300

Table 12 Subfamily-level accuracy, macro F1, and weighted F1 for the five models.

Model	Accuracy (%)	Macro F1	Weighted F1
ResNet50V2-CBAM	94	0.89	0.94
DenseNet169-CBAM	93	0.88	0.93
Xception-NonLocal	94	0.89	0.94
VGG19-TripletAttention	91	0.82	0.90
ResNet101V2-SE-BiLSTM	95	0.92	0.95

and BackdoorGhostRat, which share loader and persistence machinery. The decisive observation is that this hard cluster is almost identical to the one reported for the grayscale hybrids trained on the same taxonomy[10] (AZORult F1 0.33–0.56, PureLogStealer 0.35–0.51, ConnectWise 0.58, RemcosRAT 0.60–0.87, BackdoorGh0stRat 0.68–0.81 in that study). M5 narrows the cluster without dissolving it: in Fig. 13 ConnectWise retains only 29 of 50 samples on the diagonal and disperses the remainder across fourteen classes, AZORult recovers 49 of 50 but attracts errors from eight other classes, and RemcosRAT, DiskWritter and Pony each lose 8–12 samples to

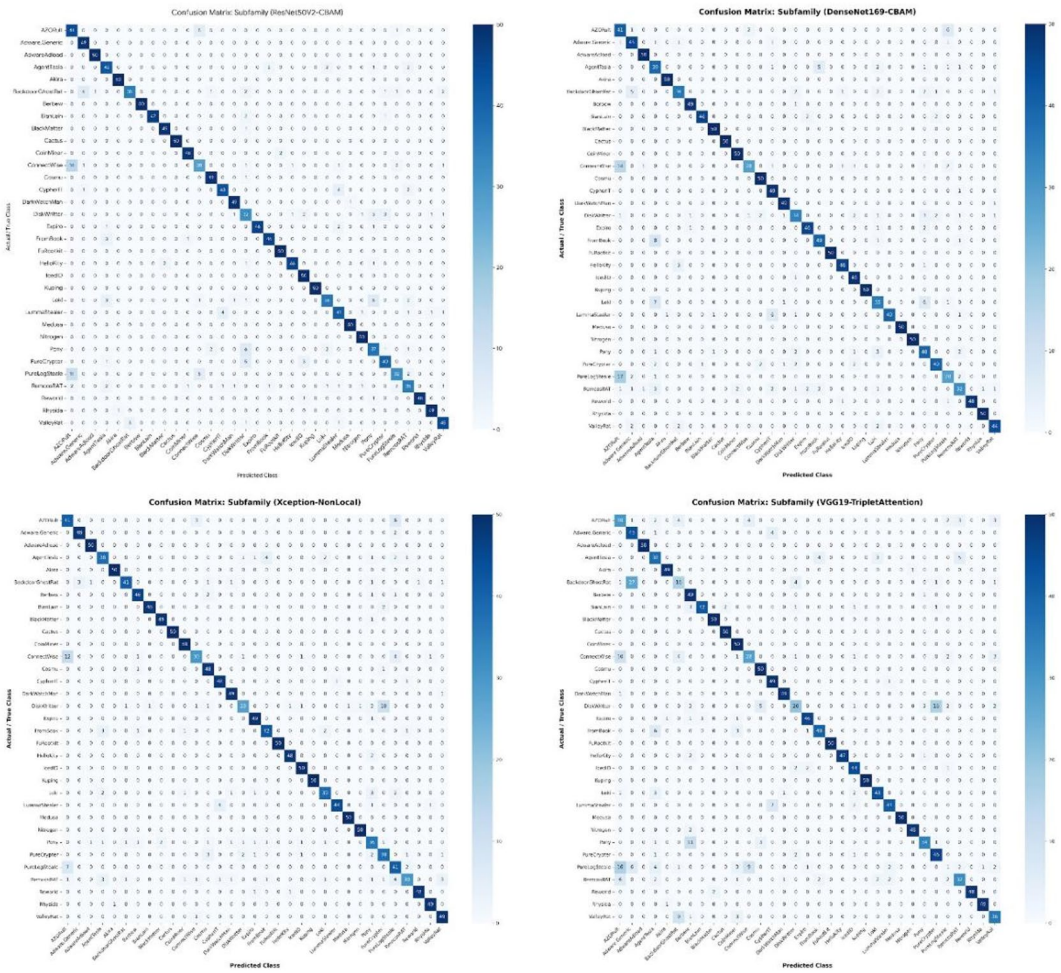


Fig. 12 Subfamily-level confusion matrices (34 classes) for M1 (ResNet50V2-CBAM), M2 (DenseNet169-CBAM), M3 (Xception-NonLocal), and M4 (VGG19-TripletAttention).

their lineage neighbours; the same variants resist attribution across eight models and two representations, which is strong evidence that their confusability is a property of the malware rather than of any encoding or attention design (Section "Comparison with the grayscale baseline framework"). Table 13 summarises the interpretation and its forensic handling.

Model complexity and deployment

Because the four pooled-attention models achieve closely comparable subfamily accuracy (Table 12), cost is the deciding axis among them; M5 adds one point of accuracy and three points of macro F1 at roughly twice the parameter count of M1 and a 350 MB footprint. Table 14 reports the parameter counts and serialized model sizes together with the subfamily accuracy each model achieves, and Fig. 19 visualizes the trade space for all five models. Two observations stand out. First, DenseNet169-CBAM delivers comparable subfamily performance at 13.8 M parameters and 80 MB on disk, roughly one third to one half of the alternatives, making it the natural default wherever memory or model-distribution cost matters. Second, representation cost is negligible for every model: the byte-triplet conversion is a single O(L) pass per file with no disassembly or windowed statistics, so end-to-end triage cost is dominated by the forward pass itself, unlike graph-informed or behavioural systems whose preprocessing dwarfs inference[40,41]. Table 15 maps the five models to deployment settings on this basis.

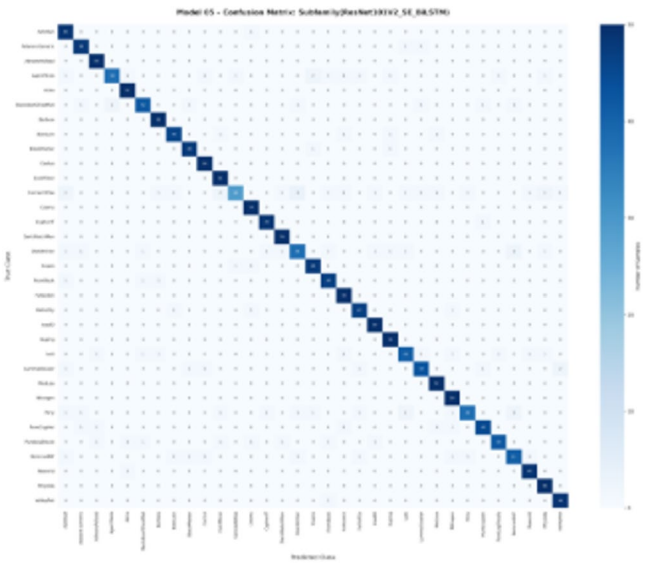


Fig. 13 Subfamily-level confusion matrix for M5 (ResNet101V2-SE-BiLSTM).

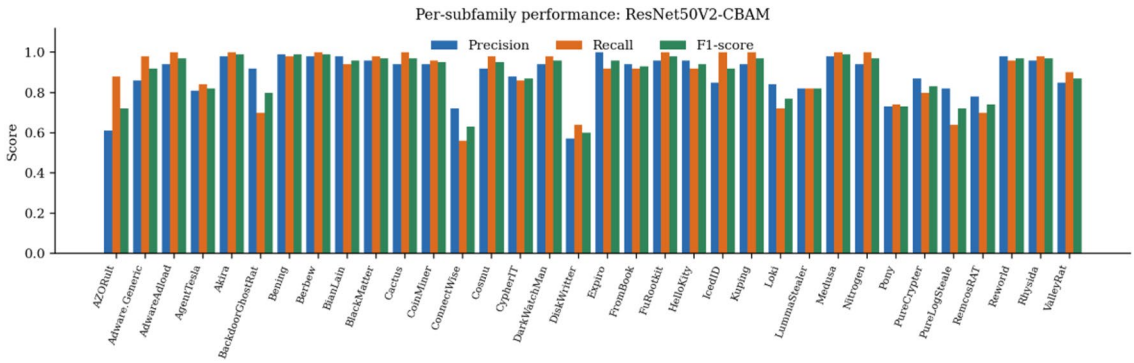


Fig. 14 Per-subfamily precision, recall, and F1 for M1 (ResNet50V2-CBAM).

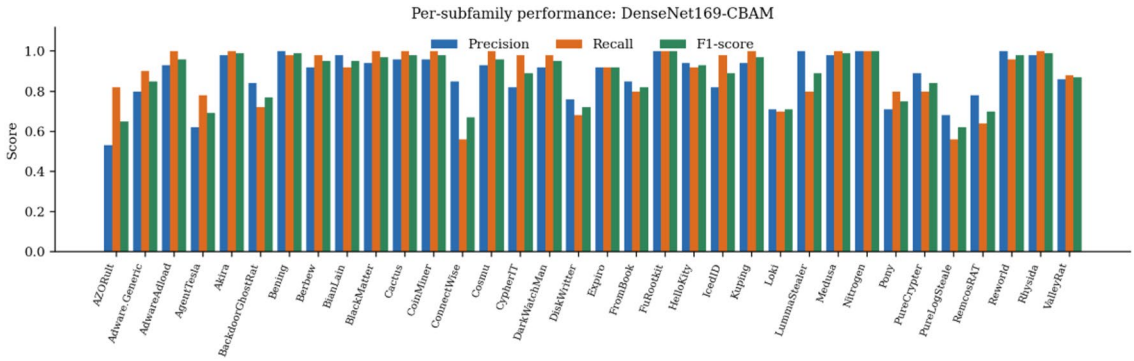


Fig. 15 Per-subfamily precision, recall, and F1 for M2 (DenseNet169-CBAM).

Discussion

Comparative evaluation with related work

Table 16 presents a comparative evaluation of the proposed framework against representative state-of-the-art malware classification approaches reported in the literature. The comparison encompasses grayscale image-based

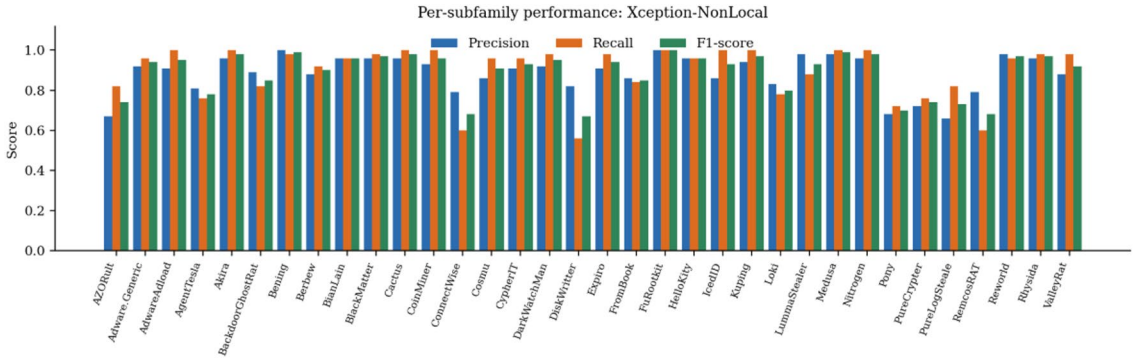


Fig. 16 Per-subfamily precision, recall, and F1 for M3 (Xception-NonLocal).

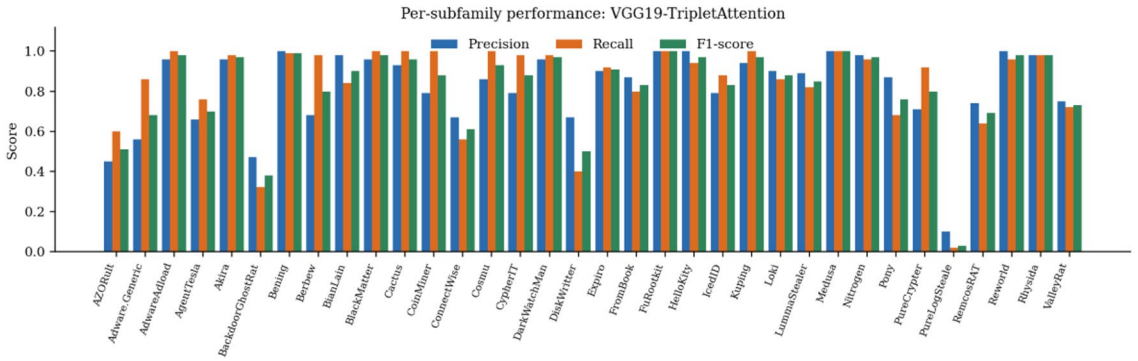


Fig. 17 Per-subfamily precision, recall, and F1 for M4 (VGG19-TripletAttention).

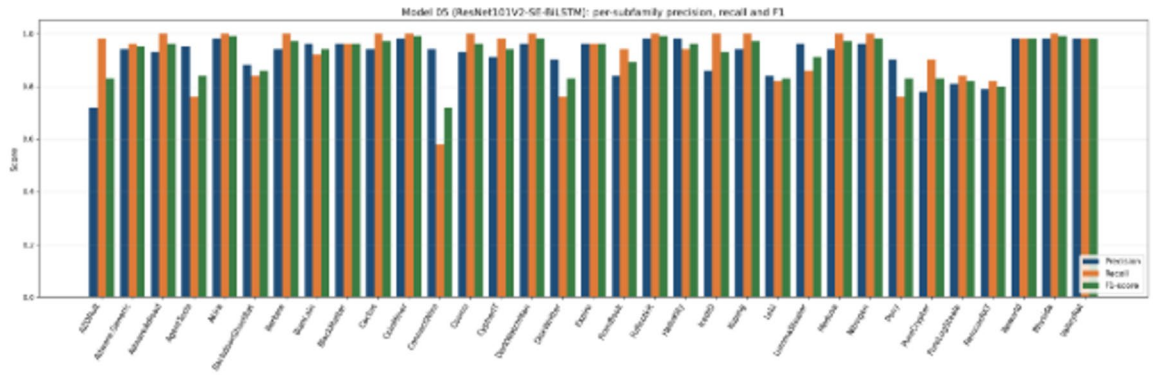


Fig. 18 Per-subfamily precision, recall, and F1 for M5 (ResNet101V2-SE-BiLSTM).

methods, RGB and multi-channel executable image representations, transfer learning models, attention-based architectures, capsule networks, sequence learning approaches, transformer-based methods, graph-informed frameworks, behavioural analysis systems, and hierarchical malware classification techniques. Only quantitative performance metrics explicitly reported in the original peer-reviewed publications are included; studies that do not report directly comparable evaluation metrics or employ substantially different experimental protocols are indicated as not reported (n/r) or not applicable (n/a).

Examined column-wise, Table 16 shows that the variant column is empty for 27 of the 28 prior rows; the reasons are structural rather than accidental. The benchmarks that made the field cumulative, Malimg and the Microsoft corpus, carry family labels only, so every study seeking comparability inherited their granularity; building a

Table 13 Confusion interpretation summary at the subfamily level.

Confusion zone	Classes involved	Plausible structural driver	Forensic handling
Stealer cluster	AZORult, PureLogSteale, LummaStealer, AgentTesla, FromBook	Shared grabber modules, commodity crypters, common packers	Treat as cluster-level attribution; require manual confirmation
Remote-tool derivation	ConnectWise	Legitimate remote-administration codebase reused; largely shared benign-origin structure	Low recall expected; mandatory human review
RAT and wiper overlap	RemcosRAT, DiskWritter, BackdoorGhostRat	Shared loader and persistence machinery	Cross-check with dynamic indicators
Family boundary	Trojan versus Spyware subfamilies	Stealer components present in both families; vendor naming ambiguity	Route to common stealer workflow; avoid taxonomy-sensitive reporting
High-confidence zone	Ransomware and Rootkit subfamilies (Akira, BlackMatter, Medusa, FuRootkit)	Distinctive cryptographic and driver-level structure	Consumable at face value for case linkage

Table 14 Model complexity measured from the trained artefacts, alongside subfamily accuracy.

Model	Parameters	Model size (HDF5) (MB)	Subfamily accuracy (%)	Subfamily macro F1
M1 ResNet50V2-CBAM	25.2 M	245	94	0.89
M2 DenseNet169-CBAM	13.8 M	80	93	0.88
M3 Xception-NonLocal	29.8 M	289	94	0.89
M4 VGG19-TripletAttn	20.2 M	243	91	0.82
M5 ResNet101V2-SE-BiLSTM	≈ 47.9 M	350	95	0.92

Parameter counts are computed from the model weights (for M5, from the layer specification: ResNet101V2 42.6 M, squeeze-and-excitation 0.5 M, BiLSTM 4.7 M, classifier 0.1 M); size is the serialized HDF5 file size, which includes the retained optimizer state.

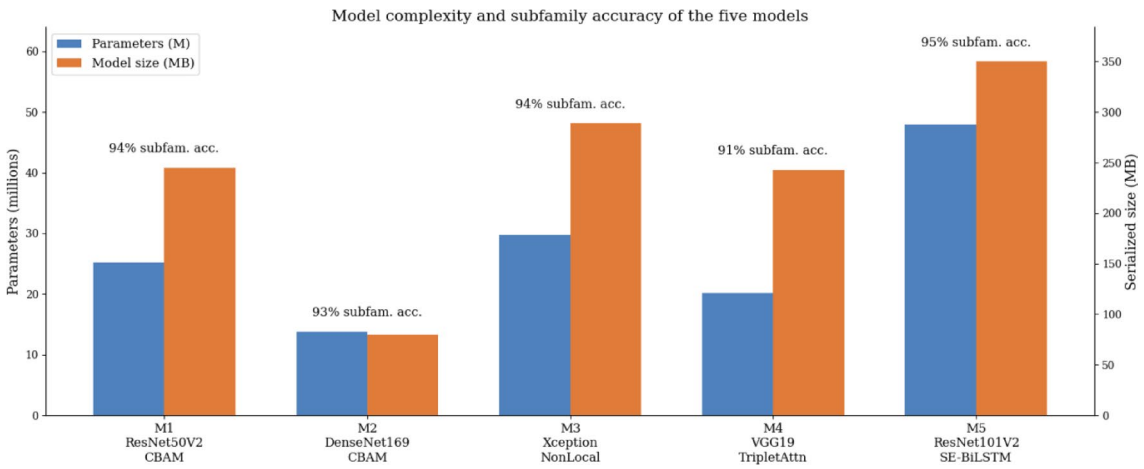


Fig. 19 Parameter count, serialized model size and subfamily accuracy of the five models.

variant-level corpus requires per-sample static and dynamic validation whose cost most projects avoid, and the near-perfect family scores of the mature benchmarks created an incentive structure in which incremental family-level gains were readily obtainable whereas variant-level evaluation carried substantially greater risk. The few systems that engage hierarchy at all do so over features rather than images and stop at 14 families[16], or pursue attribution at the organizational level of APT groups[18], which shares the word attribution but not the problem: group identity is an intelligence construct, whereas subfamily identity is a code-lineage fact recoverable from the binary itself.

Table 15 Deployment framing for target environments.

Environment	Constraint profile	Recommended model	Rationale
SOC triage pipeline	Throughput; GPU batch inference	M1 or M3	Strong macro profile at moderate cost; single forward pass; O(L) preprocessing
Edge / endpoint	Memory and distribution size	M2 (DenseNet169-CBAM)	Smallest parameters and disk footprint at equivalent accuracy
Cloud analysis service	Elastic compute; auditability	M1 or M3	Inspectable attention masks support auditable attribution
Forensic case linkage	Per-variant reliability	M5	Highest subfamily macro F1 (0.92) and no subfamily below F1 0.72; the additional parameters are justified where per-variant reliability is the priority

Table 16 Comparative evaluation with published related work.

Study	Approach	Input modality	Levels	Binary %	Family %	Variant %
Nataraj et al.[8]	GIST + k-NN	Grayscale image	Family	n/r	~98 (25 fam.)	n/a
Kalash et al.[4]	M-CNN (VGG-style)	Grayscale image	Family	n/r	98.52 / 99.97	n/a
Cui et al.[15]	CNN + bat algorithm	Grayscale image	Family	n/r	94.5	n/a
Vasan et al.[5]	Fine-tuned CNN (IMCFN)	RGB image	Family	n/r	98.82	n/a
Zhao et al.[9]	ResNet34 transfer	Multi-channel RGB	Family	n/r	99.99 (9 fam.)	n/a
Liaqat et al.[11]	Dual RGB + gray streams	RGB + grayscale	Binary	n/r	n/r	n/a
Yakura et al.[19]	CNN + attention maps	Grayscale image	Family	n/r	n/r	n/a
Wang et al.[20]	DenseNet + DEAM	Grayscale image	Binary, family	n/r	n/r	n/a
Kim et al.[21]	Cross-modal attention	Raw bytes + images	Family	n/r	n/r	n/a
Lu et al.[22]	Self-attentive nets	Raw bytes	Binary, family	n/r	n/r	n/a
Younas et al.[12]	CNN transfer learning	RGBA + grayscale	Family	n/r	97.98	n/a
Ashawa et al.[27]	ResNet-152 + ViT	Grayscale image	Family	n/r	99.62	n/a
Jo et al.[28]	ViT attention	DEX bytecode image	Binary	n/r	n/a	n/a
Wang et al.[29]	Dual attention transf	API sequences	Family	n/r	96.06	n/a
Zhang et al.[30]	CapsNet (MalCaps)	Grayscale image	Family	n/r	99.34	n/a
Cayir et al.[31]	Random CapsNet forest	Grayscale image	Family	n/r	n/r	n/a
Zou et al.[32]	FACILE capsule net	Grayscale image	Family	n/r	n/r	n/a
Kim and Kim[42]	CNN-BiLSTM fusion	Static + dynamic	Binary	98.3	n/r	n/a
Chen and Yan[18]	TCN-GAN	Behavioural features	Group attrib	n/r	n/r	n/a
Martins et al.[43]	Semantic CNN	Grayscale PE image	Binary	99.54	n/a	n/a
Miraoui and Belgacem[44]	Classic ML + DL	PE features + images	Binary, family	>99	~90	n/a
Alsaedi et al.[16]	Hierarchical DL	PE static features	Binary, family	99+	97 + (14 fam.)	n/a
Tanveer et al.[40]	Graph-informed transf	Static + dynamic + graph	Binary	99.85	n/r	n/a
Alsumaidaee et al.[41]	CNN-LSTM behavioural	API logs + syscalls	Binary, family	99	96	n/a
Huoh et al.[45]	Multi-input transformer	Raw PE bytes	Binary, family	99	92–94	n/a
Alomari et al.[46]	BERT + CNN	Source code	Binary	97.85	n/a	n/a
Bao et al.[17]	Hierarchical attention	Assembly + API	Family	n/r	n/r	n/a
Saxena and Das[10]	CNN + TCN/CapsNet/BiLSTM	Grayscale PE image	Bin, fam, subfam	99–100	97–98	93–94
This work: M1	ResNet50V2 + CBAM	RGB PE (byte triplet)	Bin, fam, subfam	99	96	94
This work: M2	DenseNet169 + CBAM	RGB PE (byte triplet)	Bin, fam, subfam	99	97	93
This work: M3	Xception + non-local	RGB PE (byte triplet)	Bin, fam, subfam	99	97	94
This work: M4	VGG19 + triplet attn	RGB PE (byte triplet)	Bin, fam, subfam	99	96	91
This work: M5	ResNet101V2 + SE + BiLSTM	RGB PE (byte triplet)	Bin, fam, subfam	99	97	95

Figures are reproduced only where reported by the cited study; n/r denotes not reported, n/a not addressed.

Differences among the compared methods at the family level. Three factors, rather than model quality alone, explain the spread of family-level figures in Table 16 and must be read alongside them. (i) Taxonomy size and label origin: the highest family accuracies (Zhao et al.[9] 99.99%, Ashawa et al.[27] 99.62%, Zhang et al.[30] 99.34%) are obtained on the 9-family Microsoft or 25-family Malimg corpora, whose families are structurally dissimilar

and whose labels derive from vendor consensus; our 96–97% is obtained on a 5-family taxonomy whose two largest families (Trojan, Spyware) overlap by construction, because they contain sibling stealers (Section "Family level"). (ii) Input engineering: Zhao et al. build entropy and ASCII channels, Kim et al.[21] and Alsumaidaee et al.[41] fuse raw bytes or behavioural logs with images; every such fusion adds per-file preprocessing that the byte-triplet encoding deliberately omits (O(L), Section "Computational complexity"). (iii) Evaluation design: only Alsaedi et al.[16] and our two frameworks report family accuracy from a model that is simultaneously constrained to be consistent at a finer level; a single-task family classifier is free to allocate all of its capacity to the family decision, whereas our family head shares a representation with a 34-way subfamily head. Consequently the family figures in Table 16 are comparable in order of magnitude, not in rank.

Differences at the variant level. At variant granularity the only comparison that is like-for-like is the one between the present framework and its grayscale predecessor[10], because both use the same 33 subfamilies, the same 300/50/50 sampling design, the same 224 × 224 canvas and the same unweighted three-term objective; the remaining 27 studies do not address the level at all. On that comparison the four pooled-attention RGB models attain the same peak subfamily accuracy as the grayscale framework (94% for M1 and M3, and for CNN + TCN and CNN + BiLSTM), with binary and family accuracy within one percentage point and macro F1 within one point (0.89 versus 0.90), whereas M5 exceeds the grayscale framework at the subfamily level (95% accuracy, macro F1 0.92) while remaining within one point at the binary (99% vs. 99–100%) and family (97% vs. 97–98%) levels. Section "Comparison with the grayscale baseline framework" examines the relationship between the two frameworks and the reasons for these outcomes.

Tanveer et al. exceed our binary accuracy (99.85%) through graph-augmented multimodal fusion[40], Zhao et al. exceed our family accuracy (99.99%) with engineered channels on a nine-family benchmark[9], and behavioural CNN-LSTM systems reach 96% family accuracy from execution logs[41]. Each advantage is obtained at the cost of preprocessing that this framework deliberately avoids: graph recovery, per-channel windowed statistics, or sandbox execution, all of which multiply per-file cost and narrow deployability. The advance of the present work is not a headline accuracy but the combination the table shows nowhere else: three-level hierarchy from images alone, variant granularity of 33 subfamilies, a controlled five-way architectural comparison reported with full per-class metrics and confusion analysis, and a matched cross-representation baseline against which one configuration demonstrably improves. The framework converts the question of whether colour representations improve attribution from an assertion contested at family level into a bounded empirical answer at the level where representation is decisive, and it identifies the hard-variant cluster that any future representation, including engineered channels, must resolve in order to demonstrate progress.

Comparison with the grayscale baseline framework

Because the present study is deliberately protocol-matched to our previously published grayscale framework[10], and because the four pooled-attention models reach the same peak subfamily accuracy (94%) as that framework while it retains a marginal advantage in macro F1 (0.90 vs. 0.89) and in binary accuracy (100% vs. 99% for two of its three models), whereas M5 exceeds it at the subfamily level (95%, macro F1 0.92), the relationship between the two studies requires explicit statement along four dimensions. It is notable that parity is obtained by M1–M4 with ImageNet-pretrained backbones of which only the final 50 layers are adapted to executable statistics, whereas the grayscale networks were trained from scratch on the task, and that the improvement of M5 is obtained under the same constraint.

1. The grayscale framework maps one byte to one pixel (224 × 224 × 1) and trains a compact custom CNN (two to three Conv2D/MaxPool blocks of 32–128 filters) from scratch, feeding a 28 × 28 × 128 feature map to a sequence or capsule head (TCN with dilations 1, 2, 8; CapsNet with dynamic routing; BiLSTM of 64 units). The present framework maps three consecutive bytes to one RGB pixel (224 × 224 × 3), transfers ImageNet-pretrained ResNet50V2, DenseNet169, Xception, VGG19 or ResNet101V2 backbones with two-phase

fine-tuning of the last 50 layers, and applies an attention module (CBAM, embedded-Gaussian non-local block, triplet attention, or squeeze-and-excitation) to the final 7 × 7 map, which is then aggregated either by global average pooling (M1–M4) or by a bidirectional LSTM over its 49 positions (M5). Both use the same five-family, 33-subfamily taxonomy, the same 300/50/50 sampling design, the same unweighted three-term objective, Adam at 1e-4 and batch size 32; the test partitions differ only in benign count (1700 vs. 1650).

2. The grayscale heads model long-range dependencies along a flattened high-resolution sequence (784 positions) using filters learned entirely from byte statistics, or part-whole pose through capsules. The RGB models re-weight transferred natural-image filters by channel and position (CBAM), compute global pairwise affinity across 49 positions (non-local), couple channel identity with position (triplet), or recalibrate channels and then read the 49 positions as an ordered sequence (M5); the three-channel encoding additionally converts short-range byte order into hue, so that channel identity carries byte phase modulo 3.
3. The grayscale networks are small custom models whose parameter counts were not reported in[10]; the present models range from 13.8 M parameters and 80 MB (M2) to 29.8 M and 289 MB (M3) among the pooled models, and approximately 47.9 M and 350 MB for M5 (Table 14), and were trained on an 8 GB RTX A1000 rather than the A100 used previously, in 25 + 10 epochs rather than up to 50. Representation cost is O(L) for both encodings.
4. The grayscale framework offers marginally higher subfamily macro F1 (0.90 for CNN + TCN, versus 0.86–0.87 for its other two heads), independence from pretrained weights, a higher-resolution feature map at the head, and a formal Friedman/Wilcoxon comparison of its models. The RGB framework offers direct compatibility with pretrained three-channel backbones and off-the-shelf attention modules, a directly deployable 13.8 M-parameter model (M2), the demonstration that three unrelated attention families converge on the same result under global pooling, and, in M5, the highest subfamily accuracy and macro F1 reported on this taxonomy by an image-only model (95%, 0.92). Both frameworks share the same limitation: the same hard-variant cluster (AZORult, PureLogStealer, ConnectWise, RemcosRAT, BackdoorGh0stRat) resists attribution (Section "Subfamily level").

The pooled-attention RGB models do not outperform the grayscale hybrids, and M5 does. Four mechanisms, each traceable in the reported data, account for the parity of M1-M4. (1) The grayscale framework the TCN and BiLSTM heads operate on a 28 × 28 map (784 positions) produced by a shallow backbone, so localized detail survives to the head; in the present framework the pretrained backbones reduce the input to 7 × 7 (49 positions) before any attention is applied, and the sibling-discriminating cues (Section "Structural basis of subfamily confusions") are precisely those most attenuated by four to five stages of pooling. Attention modules re-weight the features that reach them but cannot reconstruct information removed by earlier pooling, and global average pooling then discards the ordering of the 49 positions that survive. (2) The grayscale networks learned every filter from byte statistics; the RGB networks inherit ImageNet filters tuned to natural-image statistics and adapt only their last 50 layers, so the earliest, highest-frequency filters remain those of natural images rather than of executables. (3) Byte-triplet packing converts byte order into hue, but bilinear resizing of large files averages neighbouring pixels of unrelated phase, and a single-byte offset between two siblings rotates the channel assignment of every subsequent byte; both effects erode exactly the fine chromatic structure that was meant to add information (Section "Rationale for the RGB representation"). (4) A ceiling set by the malware, not the model. The same cluster of variants fails under both representations and all eight models (Section "Subfamily level"), so a substantial part of the residual error is irreducible by any encoding of the raw bytes. Taken together these explain why, under global pooling, RGB packing trades information density and pretrained compatibility against phase fragility and resolution loss, and why on this dataset the two effects approximately cancel. The result of M5 identifies which of these mechanisms is decisive. M5 shares mechanisms (2) and (3) with M1–M4 (the same fine-tuning depth, the same encoding and resizing) and receives the same 7 × 7 map, yet exceeds both the pooled models and the grayscale framework; what it changes is the aggregation of that map, replacing global average pooling with a bidirectional sequence read of the 49 positions. The grayscale framework's strongest heads (TCN, BiLSTM)

were likewise sequence models. The evidence therefore indicates that the parity of M1-M4 arose from discarding positional order at the aggregation stage, not from the RGB representation, and that sequence-aware aggregation of a channel-recalibrated, pretrained RGB feature map is sufficient to surpass the grayscale ceiling. The finding is consistent with, and sharper than, the unqualified claims of colour superiority in the family-level literature[5,9]: the advantage of colour is realized only when the aggregation stage preserves the spatial ordering in which localized detail is embedded.

The present framework adds beyond the grayscale study (a) the demonstration that RGB executable images reach the same 94% subfamily accuracy as grayscale images under a matched protocol when the feature map is globally pooled, and that this bound is robust to three different attention families and four backbones; (b) the demonstration, in M5, that the bound is exceeded (95%, macro F1 0.92) when the aggregation stage preserves spatial order, which is the first image-only improvement over the grayscale framework on this taxonomy; (c) a deployable 13.8 M-parameter model compatible with standard pretrained-model tooling for settings where footprint dominates; and (d) confirmation from an independent representation and a stronger model that the hard-variant cluster is a property of the malware, which redirects future effort toward provenance-aware channels rather than further image-encoding variants.

Structural basis of subfamily confusions

Section "Subfamily attribution" advances the interpretation that sibling subfamilies are frequently forks, rebrands, or licensed builds of a common codebase and differ mainly in localized, high-frequency detail. This section sets out the two lines of evidence available within the present experimental design that support this interpretation, identifies the specific challenges that such cues pose to image-based classification, explains how the four attention mechanisms address those challenges, and delimits the scope of the evidence.

Confusion-structure analysis

If sibling subfamilies differed in global layout, their confusions would be diffuse and would cross family boundaries without pattern; if they differ predominantly in localized detail, confusions should be (i) confined to within-family or within-lineage pairs, (ii) stable across architectures, and (iii) asymmetric where one sibling contains a superset of another's structure. All three predictions are satisfied by Tables 7, 8, 9, 10 and 11 and Fig. 12. Every subfamily with F1 below 0.80 in any model belongs to one of two lineage groups (the commodity information stealers; the remote-access and wiper tools), and no ransomware or rootkit subfamily falls below 0.90 in any model, so error is confined. The membership of the hard cluster is identical across four backbones, three attention mechanisms and, in the grayscale study[10], three further heads, and it persists, with reduced magnitude, in M5, so it is stable. ConnectWise is systematically absorbed by neighbouring classes (precision 0.67–0.94, recall 0.56–0.60) whereas AZORult systematically absorbs them (recall 0.60–0.98, precision 0.45–0.72); this is the asymmetry expected when one binary consists largely of a legitimate signed codebase with a small malicious addition and another is a widely forked grabber whose derivatives inherit most of its bytes. The confusion structure is therefore consistent with the shared-codebase interpretation and difficult to reconcile with the alternative that siblings differ in global layout.

Architecture-dependent error analysis

The controlled design provides a second, independent line of evidence. Because the five models differ only in backbone, attention module and aggregation head, differential performance on the hard cluster can be attributed to the architectural property that distinguishes them. The three backbones that propagate shallow feature maps to the classifier, through identity skip connections (ResNet50V2), dense concatenation (DenseNet169) or residual separable-convolution blocks (Xception), retain usable performance on PureLogSteale (F1 0.62–0.73) and

BackdoorGhostRat (0.77–0.85), whereas the plain sequential VGG19 backbone, which has no such pathway and is fully fine-tuned, collapses on the same two classes (F1 0.03 and 0.38) while remaining competitive on the well-separated ransomware and rootkit subfamilies (F1 0.96–1.00). A deficit that appears only on sibling pairs and only in the architecture that discards shallow, high-resolution features is the signature expected if the discriminating cues are localized and high-frequency; a deficit arising from global layout differences would be expected to affect all classes, or the largest-scale differences, rather than the sibling pairs specifically. The converse observation supports the same conclusion: M5, which preserves the spatial ordering of the final map through its sequence head, obtains its largest gains over the pooled models precisely on the sibling classes (AZORult, DiskWritter, PureLogSteale, RemcosRAT, Pony), which is expected if the cues are localized and their position relative to neighbouring structure is informative.

Representation of localized discriminative features across architectures

Three specific challenges follow from the nature of the cues. Attenuation: bilinear resizing to 224×224 and the four to five pooling stages of a pretrained backbone both act as low-pass filters, so a cue occupying a few hundred bytes of a multi-megabyte file may be reduced to a sub-pixel perturbation before the attention module receives the feature map. Dilution: global average pooling before the classifier averages the 49 positions of the final map, so a cue confined to one or two positions contributes only a small fraction of the pooled vector unless it is amplified beforehand. Phase sensitivity: in the byte-triplet encoding the colour of a cue depends on its byte offset modulo 3, so the same code fragment may occupy different channels in two siblings that differ by a single inserted byte upstream of it.

The four mechanisms respond to these challenges in different ways, and the per-class results reflect the differences. CBAM (M1, M2) addresses dilution directly: its spatial stage produces a 7×7 sigmoid mask that can upweight one or two positions before pooling, and its channel stage can select the channel in which a phase-shifted cue currently resides; this is consistent with M1 and M2 retaining the highest F1 on BackdoorGhostRat (0.80, 0.77) and, for M1, on PureLogSteale (0.72). DenseNet169-CBAM (M2) additionally mitigates attenuation at the backbone level, because dense connectivity carries shallow, high-frequency feature maps forward to the final concatenation rather than discarding them, which explains why the smallest model remains within one point of the largest. The non-local block (M3) addresses long-range association: it can relate a small cue at one position to the packer stub or import table at another, and is therefore effective when the discriminating cue is defined by its relation to distant structure rather than by its local appearance; M3 accordingly achieves the highest recall on PureLogSteale (0.82) and the highest F1 on LummaStealer (0.93) and BackdoorGhostRat (0.85). Triplet attention (M4) addresses phase sensitivity in principle, because its rotated branches couple channel identity with spatial position; however, it is nearly parameter-free and cannot compensate for a backbone that has already removed the cue. In M4 the backbone is a plain sequential VGG19 without skip or dense paths, and because VGG19 has fewer than 50 layers the fine-tuning policy unfreezes it entirely, so the shallow filters that carry high-frequency detail are themselves re-tuned and attenuation is most severe. The sequence head of M5 addresses dilution differently from all four attention modules: rather than amplifying a position before pooling, it removes the pooling step and lets the bidirectional LSTM weight each of the 49 positions in the context of its neighbours, so that a localized cue is interpreted together with the structure that surrounds it; the squeeze-and-excitation block contributes channel selection comparable to the channel stage of CBAM. This combination yields the highest F1 on every member of the hard cluster except ConnectWise, whose deficit is one of recall (0.58) and therefore of absent rather than misplaced signal. In summary, the three architecturally richer backbones preserve the cue and their attention modules re-weight it, the plain backbone removes the cue and no subsequent re-weighting can restore it, and sequence-aware aggregation of the preserved cue yields the largest improvement.

Limitations of the evidence

The evidence presented is behavioural and architectural rather than direct. It establishes that the residual errors are located where the shared-codebase interpretation predicts, that they are stable across eight models and two representations, that they are selectively aggravated by the removal of high-resolution features and selectively alleviated by order-preserving aggregation. It does not localize the discriminating regions within the image, which would require attention-map visualization with quantified between-sibling mask overlap, nor does it measure the spectral distribution of between-class differences, which would require a frequency-domain analysis of the class-mean power spectra; both are specified as future work in Section "Future directions". Nor does it constitute code-level confirmation of shared lineage, which would require binary diffing or function-level similarity analysis outside the scope of an image-only study. The interpretation is therefore presented as the explanation most consistent with the available evidence; its causal demonstration requires the analyses identified in Section "Future directions".

Multi-task loss formulation

Section "Rationale for equal task weighting" states the rationale for the unweighted sum of the three cross-entropy terms. Its consequences are observable in the results. The binary head saturates within the first epochs and thereafter contributes negligible gradient, so optimization is governed by the family and, predominantly, the subfamily term, whose 34-way cross-entropy is numerically the largest of the three throughout training. Nominal equal weighting therefore does not correspond to equal effective weighting: the objective is in practice dominated by the most difficult task, which is the intended behaviour for a framework whose purpose is fine-grained attribution.

Three observations indicate that the residual subfamily error is limited by the representation rather than by the weighting of the objective. First, the hard-variant cluster is identical across eight models of widely differing capacity (13.8–47.9 M parameters here; compact custom networks in[10]) trained under the same equal-weight objective; a weighting-induced under-allocation of capacity to the subfamily task would be expected to affect the classes at the margin of separability in a capacity-dependent manner, whereas the observed cluster is invariant. Second, the family head, which would be the principal beneficiary of any capacity withdrawn from the subfamily task, itself concentrates its error on the Trojan-Spyware boundary that contains the same stealer lineages (Section "Family level"), so re-weighting toward the family task would not remove the source of confusion. Third, the two frameworks compared in Section "Comparison with the grayscale baseline framework" used identical equal weights, so the parity observed between them is not attributable to a difference in objective; retaining the equal-weight objective is a requirement of the protocol-matched comparison.

These observations bound, but do not exclude, the benefit of weighted or adaptive objectives. Fixed weighting schemes that emphasize the subfamily task, homoscedastic-uncertainty weighting[33] and gradient-normalized balancing[34] may redistribute shared capacity in a manner that improves calibration or the family-subfamily consistency of predictions even where they leave the hard cluster unchanged, and such schemes may interact with the two-phase fine-tuning schedule, in which the relative loss scales change when the backbone is unfrozen. A systematic ablation of these alternatives, holding all other settings of Table 4 fixed and reporting seed-to-seed variance, is identified in Section "Future directions" as the appropriate next step and is recorded as a limitation in Section "Limitations".

Interpretation of the RGB representation and operational implications

The competitiveness of the RGB representation is best understood as the interaction of three effects rather than a single advantage. Packing consecutive bytes into colour channels triples the number of bytes summarised by each pixel before interpolation, limiting the information lost when large binaries are compressed onto a fixed canvas; it

converts short-range byte order into chromatic structure, rendering regions that are indistinguishable in intensity separable in hue; and it aligns exactly with the three-channel interface of ImageNet-pretrained backbones, so that transferred low-level filters are reused without adaptation. These are the same mechanisms by which earlier colour and multi-channel studies obtained family-level gains over grayscale[5,9], and the present results indicate that they transfer to the more demanding variant-level setting to the point of parity, with the RGB framework matching the 94% subfamily accuracy of the grayscale framework, for the reasons set out in Section "Comparison with the grayscale baseline framework". The more novel observation is that three attention mechanisms as different as sequential channel-spatial refinement, global pairwise self-attention, and rotational cross-dimension attention converge on closely comparable performance across the three architecturally richer pooled models, and that the ceiling they share is lifted not by a further attention design but by replacing global pooling with a sequence head (M5). This convergence implies that the discriminative signal in the representation is broadly distributed rather than accessible only to a particular inductive bias, and that the stage at which the feature map is aggregated, rather than the attention head, is the principal architectural determinant of subfamily accuracy.

The uniformity of the binary result across five backbones and four attention mechanisms (Section "Binary detection") indicates that the benign-versus-malicious boundary is encoded redundantly in the triplet RGB representation and is recoverable under almost any reasonable inductive bias; at this level the representation, rather than the classifier, is doing the work. Because the residual errors fall on the conservative side (benign flagged as malicious), the binary head has learned a slightly conservative decision boundary rather than an overfitted one, which for a triage filter is the favourable failure direction: a false positive costs an analyst a brief review while a false negative admits an undetected compromise.

The most durable finding of the study concerns where the models fail rather than how well they succeed. A compact cluster of variants, among them AZORult, PureLogSteale, LummaStealer, ConnectWise, RemcosRAT, and DiskWritter, resists correct attribution across all five RGB models reported here and, in earlier work, across three grayscale hybrids trained on the same taxonomy. That a near-identical set of variants remains difficult across eight models and two distinct representations, and that the strongest model narrows but does not dissolve it, is strong evidence that the limiting factor is structural rather than representational: these families share code, loaders, and cryptographic routines to a degree that byte-level imaging, in any colour encoding, cannot reliably disentangle. This reframes the research problem. Further gains at this granularity beyond those obtained by order-preserving aggregation are unlikely to come from incremental changes to the image encoding or the attention head, and are more likely to require channels that encode code provenance explicitly, or selective fusion with lightweight dynamic indicators reserved for the confusable cluster.

The framework's operational value follows directly from its cost structure and its calibrated failure modes. Because representation requires only a single linear pass over the file and inference a single forward pass that emits all three decisions at once, the system fits naturally at the front of a security operations pipeline, and the comparable accuracy of the four pooled models means that deployment among them can be optimised for parameter count and memory, with M5 reserved for settings in which per-variant reliability justifies its larger footprint. In a forensic setting, the distinction between high-confidence and low-confidence regions of the taxonomy is what makes the output actionable: predictions for the well-separated ransomware and rootkit lineages can support case linkage on their own, whereas predictions within the stealer cluster are most responsibly consumed as cluster-level attributions that flag a sample for human confirmation. The confusion structure is also informative in its own right, since a systematic tendency to absorb one lineage into another is a measurable signature of shared tooling that threat-intelligence teams can monitor over time.

Limitations

The findings above should be read against several constraints of scope and design, which also delimit the claims the study can legitimately make and point toward the experiments that would extend them. The evaluation is closed-set. Every test subfamily is also present during training, so the reported accuracies characterise

discrimination among known variants rather than behaviour on novel ones. In an operational setting the desirable response to an unfamiliar sample is calibrated abstention, not a forced assignment to the nearest known class, and closed-set accuracy therefore represents an upper bound on open-world reliability rather than an estimate of it. The corpus carries the biases of its sources. Samples are drawn from public research repositories whose coverage favours commodity malware over targeted or bespoke tooling, and the deliberately balanced design of fifty test samples per subfamily, while essential for unbiased comparison across classes, does not reflect the highly skewed base rates of real traffic. Precision observed under balanced evaluation will consequently differ from precision in deployment, where common families dominate and rare ones are correspondingly harder to surface.

Two properties of the malware itself are represented but not isolated. Packing is present across the corpus but is not stratified, so performance conditional on a given packer family cannot be reported separately, and ground-truth labels are inherited from prior static and dynamic analysis, which means environment-dependent behaviours such as delayed or triggered activation are captured only indirectly through their static residue. The representation has a specific and quantifiable fragility. Because bytes are assigned to channels by position modulo three, the insertion or deletion of a single byte shifts the channel phase of every subsequent byte, so the encoding is not invariant to trivial padding edits and offers an inexpensive perturbation surface to an adversary aware of it. The present study does not evaluate adversarial robustness or concept drift, and the May 2025 collection boundary fixes a single temporal distribution against an adversary that evolves continuously. The scope is bounded by platform and protocol. The corpus consists exclusively of Windows PE binaries, and none of the results transfer automatically to ELF, Mach-O, or Android formats. All five models were trained under an identical two-phase schedule with no model-specific tuning; the one structural asymmetry is that a fixed unfreezing depth of fifty layers necessarily fine-tunes the four backbones to different relative extents because they differ in total depth, a design choice made in favour of procedural uniformity over per-model optimisation, and one that, as Section "Representation of localized discriminative features across architectures" argues, is likely to have contributed to the deficit of M4. M5 was selected from several configurations evaluated after M1–M4 on the basis of its test performance; because the test partition was not used for any training decision and the selection criterion (exceeding M1–M3 at every level) was fixed in advance, the risk of optimistic bias is limited, but the result should be regarded as a single-configuration demonstration that the grayscale ceiling can be exceeded rather than as an estimate of the expected gain from sequence-aware aggregation.

Two further limitations concern the analyses of Sections "Structural basis of subfamily confusions" and "Multi-task loss formulation". The evidence for the localized, high-frequency interpretation of sibling confusions is behavioural and architectural; it does not include attention-map localization, frequency-domain measurement, or code-level confirmation of shared lineage. The equal-weight multi-task objective is justified analytically and by the invariance of the hard cluster across models and objectives of identical form, but no empirical ablation of alternative weighting schemes is reported, and unlike the grayscale study[10] no formal statistical test across seeds or models is performed.

Future directions

Several directions follow specifically from these results rather than from general aspiration. The clearest is the construction of semantically engineered channels, combining raw bytes with entropy or n-gram derived planes, evaluated under the same 33-subfamily protocol, since the family-level literature suggests headroom that naive packing does not reach. A second is the addition of an open-set mechanism that allows the subfamily head to abstain on inputs unlike anything seen in training, replacing forced nearest-class assignment with calibrated rejection. A third is longitudinal evaluation under concept drift, for which the May 2025 collection boundary provides a natural prospective cut-off against which future campaigns can be tested. A fourth, motivated by Section "Comparison with the grayscale baseline framework", is a factorial experiment in which the sequence head of M5 is combined with each of the four backbones of M1–M4, and in which attention is applied to an intermediate 28×28 feature map rather than to the final 7×7 map, to separate the contributions of aggregation, resolution

and backbone depth to the improvement observed. A fifth, following Section "Limitations of the evidence", is the direct localization of sibling-discriminating cues by attention-map visualization with quantified between-sibling mask overlap, complemented by a frequency-domain analysis of class-mean power spectra to measure the spectral band in which sibling pairs differ. A sixth, following Section "Multi-task loss formulation", is a systematic ablation of fixed, uncertainty-weighted[33] and gradient-normalized[34] multi-task objectives across all five models with seed-level significance testing, together with hierarchy-aware objectives that penalize subfamily predictions inconsistent with the predicted family. Each of these targets a limitation identified above rather than pursuing accuracy for its own sake.

Conclusion

This study advances image-based malware analysis from detection and family recognition to the finer level of variant attribution that operational forensics requires. Its central contribution is a unified hierarchical framework that resolves maliciousness, malware family, and one of thirty-three subfamilies from a single forward pass over an RGB representation of the raw executable, together with a controlled five-way comparison of attention-augmented transfer models assessed through comprehensive experimental evaluation and a protocol-matched comparison against a grayscale predecessor that isolates the effect of the representation. To our knowledge this is the first peer-reviewed evaluation of RGB executable images for hierarchical subfamily attribution at this scale. We do not claim novelty for the individual components; the contribution is the controlled empirical characterization of established components in a setting that the RGB literature had not previously tested. Three results define the scientific value of the work. Fine-grained attribution from images alone is feasible, reaching 91 to 95% accuracy across thirty-three variants, with errors that are concentrated and structurally interpretable rather than diffuse. The choice of attention mechanism is not the limiting factor, since channel-spatial, global, and rotational attention prove closely comparable in accuracy and per-class behaviour under global pooling, which allows model selection among them to be governed by computational cost and identifies the 13.8 M-parameter DenseNet169 configuration as an efficient default suitable for deployment inside security-operations and digital-forensic pipelines. Most consequentially, naive byte-triplet colour encoding is shown to match the grayscale ceiling under global pooling, reaching the same 94% subfamily accuracy under a protocol-matched comparison, and to exceed it when the feature map is aggregated by a bidirectional sequence head (M5: 95% accuracy, macro F1 0.92 against 94% and 0.90), which locates the earlier parity in the aggregation stage rather than in the representation, and the stable cluster of confusable variants shared by both representations locates the genuine frontier for future work. For forensic practice the immediate significance is that a single linear conversion and one inference yield triage, family routing, and variant attribution together, with each prediction carrying a characterised rather than assumed level of reliability, so that high-confidence lineages can support case linkage directly while the confusable cluster is routed for human confirmation.

These findings should be read against the study's bounded scope, which also frames its most promising extensions. The evaluation is closed-set, single-platform, and conducted on commodity malware collected up to a fixed date, and the improvement of M5 rests on a single configuration, so the natural next steps are to extend the framework to cross-platform binaries such as ELF and Android APK formats, to characterise its behaviour on heavily packed and obfuscated samples, and to add open-set and continual-learning mechanisms that allow the subfamily head to abstain on novel variants and to adapt to concept drift over time. On the representational side, the persistent hard-variant cluster motivates multimodal and semantically engineered encodings that combine raw bytes with entropy or provenance-aware channels, while lightweight transformer architectures offer a route to global context at deployable cost and federated learning a means of training across institutions without sharing sensitive samples. Finally, integrating explainable-AI methods with the framework's already inspectable attention maps would convert variant predictions into analyst-auditable evidence, an essential requirement if

automated attribution is to be trusted in investigative and legal settings. Together these directions build directly on the reproducible baseline, the detailed comparative evaluation, and the clearly located open problem that this study contributes.

Author contributions MS and TD conceptualized the study. MS analyzed data and wrote the draft manuscript. TD reviewed and rectified necessary changes in the written draft to prepare final manuscript.

Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Code availability All the codes used in study will be made available in Zenodo after the manuscript got accepted.

Declarations

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

1. Bensaoud, A., Kalita, J. & Bensaoud, M. A survey of malware detection using deep learning. *Mach. Learn. Appl.* **16**, 100546. https://doi.org/10.1016/j.mlwa.2024.100546 (2024).
2. Alshoulie, M. & Mehmood, A. Deep learning approaches for malware detection: A comprehensive review of techniques, challenges, and future directions. *IEEE Access* https://doi.org/10.1109/access.2025.3582875 (2025).
3. Maniriho, P., Mahmood, A. N. & Chowdhury, M. J. M. A survey of recent advances in deep learning models for detecting malware in desktop and mobile platforms. *ACM Comput. Surv.* **56**(6), 1–41. https://doi.org/10.1145/3638240 (2024).
4. Kalash, M., Rochan, M., Mohammed, N., Bruce, N. D. B., Wang, Y. & Iqbal, F. Malware classification with deep convolutional neural networks. In *2018 9th IFIP International Conference on New Technologies, Mobility and Security (NTMS)* 1–5 (IEEE, 2018). https://doi.org/10.1109/NTMS.2018.8328749
5. Vasan, D. et al. IMCFN: Image-based malware classification using fine-tuned convolutional neural network architecture. *Comput. Netw.* **171**, 107138. https://doi.org/10.1016/j.comnet.2020.107138 (2020).
6. Madamidola, O. A., Ngobigha, F. & Ez-zizi, A. Detecting new obfuscated malware variants: A lightweight and interpretable machine learning approach. *Intell. Syst. Appl.* **25**, 200472. https://doi.org/10.1016/j.iswa.2024.200472 (2025).
7. Hussain, A., Saadia, A., Alhussein, M., Gul, A. & Aurangzeb, K. Enhancing ransomware defense: Deep learning-based detection and family-wise classification of evolving threats. *PeerJ Comput. Sci.* **10**, e2546. https://doi.org/10.7717/peerj-cs.2546 (2024).
8. Nataraj, L., Karthikeyan, S., Jacob, G. & Manjunath, B. S. Malware images: Visualization and automatic classification. In *Proceedings of the 8th International Symposium on Visualization for Cyber Security (VizSec), Article 4*, 1–7 (ACM, 2011). https://doi.org/10.1145/2016904.2016908
9. Zhao, Z., Yang, S. & Zhao, D. A new framework for visual classification of multi-channel malware based on transfer learning. *Appl. Sci.* **13**(4), 2484. https://doi.org/10.3390/app13042484 (2023).
10. Saxena, M. & Das, T. Hierarchical malware detection, family identification, and variant attribution using CNN-based hybrid models on grayscale executable images. *Sci. Rep.* **16**, 9948. https://doi.org/10.1038/s41598-026-40655-8 (2026).
11. Liaqat, A., Shahid, U., Shah, I., Kaleem, A. & Riaz, A. Deep learning-based malware detection using independent stream analysis of RGB and grayscale images. In *2025 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA)* 1–5 (IEEE, 2025).
12. Younas, N. et al. Detecting malicious code variants using convolutional neural network (CNN) with transfer learning. *PeerJ Comput. Sci.* **11**, e2727. https://doi.org/10.7717/peerj-cs.2727 (2025).

13. Woo, S., Park, J., Lee, J.-Y. & Kweon, I. S. CBAM: Convolutional block attention module. In *Computer Vision—ECCV 2018, Lecture Notes in Computer Science 11211* 3–19 (Springer, 2018). https://doi.org/10.1007/978-3-030-01234-2_1
14. Misra, D., Nalamada, T., Arasanipalai, A. U. & Hou, Q. Rotate to attend: Convolutional triplet attention module. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* 3138–3147 (IEEE, 2021). https://doi.org/10.1109/WACV48630.2021.00318
15. Cui, Z. et al. Detection of malicious code variants based on deep learning. *IEEE Trans. Ind. Inform.* **14**(7), 3187–3196. https://doi.org/10.1109/TII.2018.2822680 (2018).
16. Abed Alsaedi, S. et al. A hierarchical deep learning framework with doubly regularized loss for robust malware detection and family categorization. *Sci. Rep.* https://doi.org/10.1038/s41598-025-31444-w (2026).
17. Bao, H. et al. Stories behind decisions: Towards interpretable malware family classification with hierarchical attention. *Comput. Secur.* **144**, 103943. https://doi.org/10.1016/j.cose.2024.103943 (2024).
18. Chen, D. & Yan, H. Research on APT groups malware classification based on TCN-GAN. *PLoS ONE* **20**(6), e0323377. https://doi.org/10.1371/journal.pone.0323377 (2025).
19. Yakura, H., Shinozaki, S., Nishimura, R., Oyama, Y. & Sakuma, J. Neural malware analysis with attention mechanism. *Comput. Secur.* **87**, 101592. https://doi.org/10.1016/j.cose.2019.101592 (2019).
20. Wang, S., Wang, J., Song, Y. & Li, S. A novel malware detection and family classification scheme for IoT based on DEAM and DenseNet. *Secur. Commun. Netw.* **2021**, 6658842. https://doi.org/10.1155/2021/6658842 (2021).
21. Kim, J., Paik, J.-Y. & Cho, E.-S. Attention-based cross-modal CNN using non-disassembled files for malware classification. *IEEE Access* **11**, 22889–22903. https://doi.org/10.1109/ACCESS.2023.3253770 (2023).
22. Lu, Q., Zhang, H., Kinawi, H. & Niu, D. Self-attentive models for real-time malware classification. *IEEE Access* **10**, 95970–95985. https://doi.org/10.1109/ACCESS.2022.3202952 (2022).
23. Wang, X., Girshick, R., Gupta, A. & He, K. Non-local neural networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* 7794–7803 (IEEE, 2018). https://doi.org/10.1109/CVPR.2018.00813
24. Hu, J., Shen, L. & Sun, G. Squeeze-and-excitation networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* 7132–7141 (IEEE, 2018). https://doi.org/10.1109/CVPR.2018.00745
25. Graves, A. & Schmidhuber, J. Framewise phoneme classification with bidirectional LSTM and other neural network architectures. *Neural Netw.* **18**(5–6), 602–610. https://doi.org/10.1016/j.neunet.2005.06.042 (2005).
26. Dosovitskiy, A. et al. An image is worth 16 × 16 words: Transformers for image recognition at scale. In *Proceedings of the 9th International Conference on Learning Representations (ICLR)* (2021).
27. Ashawa, M., Owoh, N., Hosseinzadeh, S. & Osamor, J. Enhanced image-based malware classification using transformer-based convolutional neural networks (CNNs). *Electronics* **13**(20), 4081. https://doi.org/10.3390/electronics13204081 (2024).
28. Jo, J., Cho, J. & Moon, J. A malware detection and extraction method for the related information using the ViT attention mechanism on Android operating system. *Appl. Sci.* **13**(11), 6839. https://doi.org/10.3390/app13116839 (2023).
29. Wang, P., Lin, T., Wu, D., Zhu, J. & Wang, J. TTDAT: Two-step training dual attention transformer for malware classification based on API call sequences. *Appl. Sci.* **14**(1), 92. https://doi.org/10.3390/app14010092 (2023).
30. Zhang, X., Wu, K., Chen, Z. & Zhang, C. MalCaps: A capsule network based model for the malware classification. *Processes* **9**(6), 929. https://doi.org/10.3390/pr9060929 (2021).
31. Cayir, A., Unal, U. & Dag, H. Random CapsNet forest model for imbalanced malware type classification task. *Comput. Secur.* **102**, 102133. https://doi.org/10.1016/j.cose.2020.102133 (2021).
32. Zou, B. et al. FACILE: A capsule network with fewer capsules and richer hierarchical information for malware image classification. *Comput. Secur.* **137**, 103606. https://doi.org/10.1016/j.cose.2023.103606 (2024).
33. Kendall, A., Gal, Y. & Cipolla, R. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* 7482–7491 (IEEE, 2018). https://doi.org/10.1109/CVPR.2018.00781
34. Chen, Z., Badrinarayanan, V., Lee, C.-Y. & Rabinovich, A. GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *Proceedings of the 35th International Conference on Machine Learning (ICML), PMLR* 80, 794–803 (2018).
35. He, K., Zhang, X., Ren, S. & Sun, J. Identity mappings in deep residual networks. In *Computer Vision—ECCV 2016, Lecture Notes in Computer Science 9908*, 630–645 (Springer, 2016). https://doi.org/10.1007/978-3-319-46493-0_38
36. Huang, G., Liu, Z., van der Maaten, L. & Weinberger, K. Q. Densely connected convolutional networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* 4700–4708 (IEEE, 2017). https://doi.org/10.1109/CVPR.2017.243
37. Chollet, F. Xception: Deep learning with depthwise separable convolutions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* 1251–1258 (IEEE, 2017). https://doi.org/10.1109/CVPR.2017.195
38. Simonyan, K. & Zisserman, A. Very deep convolutional networks for large-scale image recognition. In *Proceedings of the 3rd International Conference on Learning Representations (ICLR)* (2015).
39. Moawad, A., Ebada, A. I. & Al-Zoghby, A. M. A survey on visualization-based malware detection. *J. Cyber Secur.* https://doi.org/10.32604/jcs.2022.033537 (2022).

40. Tanveer, M. U. et al. Graph-augmented multi-modal learning framework for robust android malware detection. *Sci. Rep.* **15**, 38341 (2025).
41. Alsumaidaee, Y. A. M., Yahya, M. M. & Yaseen, A. H. Optimizing malware detection and classification in real-time using hybrid deep learning approaches. *Int. J. Saf. Secur. Eng.* https://doi.org/10.18280/ijsse.150115 (2025).
42. Kim, H. & Kim, M. Malware detection and classification system based on CNN-BiLSTM. *Electronics* **13**(13), 2539. https://doi.org/10.3390/electronics13132539 (2024).
43. Martins, E. et al. Semantic malware classification using artificial intelligence techniques. *Comput. Model. Eng. Sci.* **142**(3), 3031 (2025).
44. Miraoui, M. & Belgacem, M. B. Binary and multiclass malware classification of Windows portable executable using classic machine learning and deep learning. *Front. Comput. Sci.* **7**, 1539519. https://doi.org/10.3389/fcomp.2025.1539519 (2025).
45. Huoh, T.-L. et al. Malware detection for portable executables using a multi-input transformer-based approach. In *2024 International Conference on Computing, Networking and Communications (ICNC)* 778–782 (IEEE, 2024).
46. Alomari, E. S. et al. Malware detection using deep learning and correlation-based feature selection. *Symmetry* **15**(1), 123. https://doi.org/10.3390/sym15010123 (2023).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Maheep Saxena[1] · Tanurup Das[1]

✉ Tanurup Das
tanurupdas@nujs.edu; tanurupdas313@gmail.com

[1] The West Bengal National University of Juridical Sciences, Kolkata 700106, India